King Saud University
College of Computer and Information Sciences
Department of Information Technology

IT 320 Course Project
Semester-2, 1447H

Software Product Release

<رشيد>

Prepared by

<Shatha bin mana, 445202291>

<Tala Alqahtani, 445204552>

<Lama Almubarak, 445200338>

<Jood Alkhneen, 445201348>

Supervised by

<Reem F. Alqifari >

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1 Chapter 1: Introduction

Sustainable management of natural resources has become a critical concern, particularly in sectors related to water and electricity consumption. Rapid urban development, population growth, and increasing dependence on infrastructure systems have placed growing pressure on these essential resources. In Saudi Arabia, where natural freshwater resources are limited and energy demand is high, improving the efficiency and sustainability of water and electricity management is of significant national importance.

Water resource management in the Kingdom is a strategic priority due to heavy reliance on desalination, which carries substantial environmental and economic costs. The Saudi National Water Strategy 2030 emphasizes the need to reform the water and wastewater sector to ensure the sustainable development of water resources while maintaining high-quality and affordable services.[1] This highlights the importance of improving monitoring, coordination, and management practices related to water and electricity usage.

Technology plays a key role in supporting these sustainability efforts. Digital platforms can enhance communication between communities and service providers, enable structured data collection, and support informed decision-making. By improving visibility into resource-related issues and facilitating timely responses, technological solutions contribute to reducing waste, improving operational efficiency, and supporting long-term infrastructure planning at both local and national levels.

This proposal outlines the problem addressed by the project, the proposed solution, the product vision and roadmap, project objectives, scope, Scrum team roles, and supporting references. The following section discusses the problem in detail, focusing on the limitations of current practices and the need for a more effective approach.

1.1 The Problem

Saudi Arabia faces increasing pressure on its water and electricity infrastructure due to urban growth and rising service demand. Efficient management of utility-related issues is especially critical in a country where water is a scarce resource. Reports indicate that water losses in Saudi Arabia's distribution networks range between 25–40% of total supply due to leaks, aging infrastructure, and operational inefficiencies [2]. This significant loss highlights the importance of early detection and effective reporting of utility-related problems to reduce unnecessary waste and improve service reliability.

At the community level, Saudi Arabian residents and individuals living within the Kingdom frequently encounter issues such as water leaks, electricity outages, and malfunctioning equipment. However, the process of reporting these problems is often fragmented and inconsistent. While some digital platforms do exist, there remains an opportunity to improve how reports are tracked, prioritized, and communicated. In some cases, reporting processes may even rely on phone-based communication or multiple informal channels, which can lead to inconsistent submissions, limited follow-up visibility, or uncertainty regarding case progress.

For example, a resident who notices a water leak in a neighborhood street may attempt to report it through multiple informal channels. Without a coordinated reporting mechanism, the report may lack precise location details, supporting evidence, or priority classification. During this time, the leak may continue, contributing to resource waste and prolonged disruption.

From a system perspective, the absence of a unified and structured reporting approach results in fragmented data collection, limited case traceability, and reduced transparency in issue resolution. Reports received through informal methods are difficult to categorize, prioritize, and monitor efficiently, limiting the ability of responsible entities to manage cases in a consistent and organized manner.

Therefore, there is a clear need to improve how utility-related issues are reported and tracked for residents in Saudi Arabia in order to enhance communication, organization, and transparency between community members and responsible entities. The following section presents the proposed solution designed to address this reporting gap.

1.2 The Solution

This section presents the proposed solution, Rasheed, a web-based system designed to simplify the process of reporting water and electricity issues within the community. Instead of relying on phone calls or informal messages, users can report problems such as water leaks, electricity outages, or malfunctioning equipment through a single centralized online platform.

Users submit reports through a simple form where they describe the issue and provide basic location details, including the neighborhood name, street name, and building number. These details help identify the location of the problem without the need for maps or advanced location tools.

The system also allows users to attach an image of the issue to better illustrate the situation. In addition, users select a severity level to indicate the urgency of the problem, which supports more effective prioritization and handling of reports.

All submitted reports are stored in a centralized system, making it easier to review, manage, and track reported problems. This helps reduce confusion and minimizes duplicate reporting of the same issue.

To encourage community participation, Rasheed includes a simple reward mechanism. Users earn 10 points for each verified report, and once they accumulate 100 points through 10 valid reports, they receive a badge that recognizes their contribution to conserving water and electricity resources.

Overall, Rasheed provides a practical and user-friendly approach to improving issue reporting while promoting community engagement and supporting the sustainable use of water and electricity resources. Based on this solution, the next section introduces the Product Vision of Rasheed, which outlines the overall purpose of the system and the key goals it aims to achieve.

1.3 The Product

1.3.1 Product Vision

For community members and local residents in Saudi Arabia

Who seek to support environmental sustainability and reduce water and electricity waste

The Rasheed is a web-based sustainability reporting platform

That enables users to report water and electricity issues and contribute to environmental protection

Unlike Balady

Our product promotes sustainable behavior by rewarding users with points and badges for verified reports.

1.3.2 Product Roadmap

In this section, we present the product roadmap for the Rasheed platform, which illustrates the development process across the project phases demonstrates how the system will be built and delivered over time.

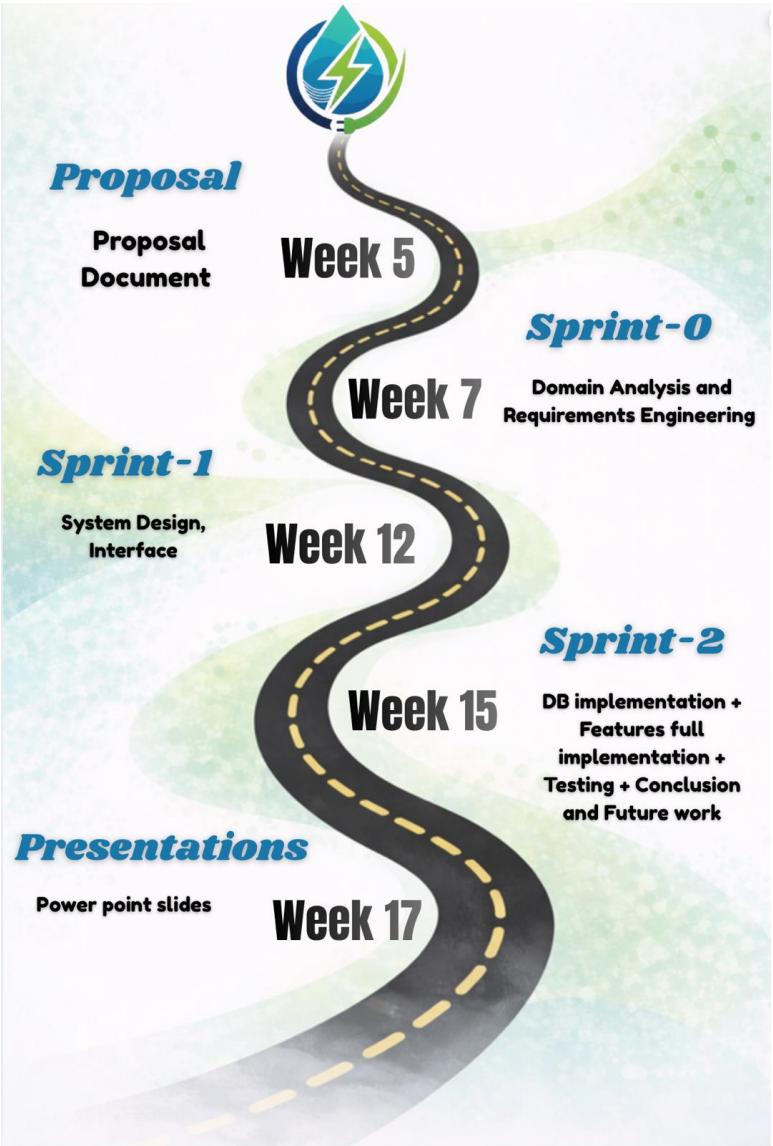


Figure 1: Roadmap

1.3.3 Objectives

This section explains the main goals of the Rasheed project. The objectives are divided into product goals, project goals, and learning goals.

Product Objectives (customer focus-value):

These objectives are related to the product itself, including problems we aim to solve and benefits and features we aim to provide the user with. Main benefits are:

- Provide user registration functionality to allow users to create accounts.
- Provide login functionality to allow users to access their accounts.
- Provide logout functionality to allow users to exit their accounts.
- Provide the ability for users to add new reports.
- Provide the ability for users to edit their reports.
- Provide the ability for users to delete their reports.
- Provide search functionality to help users find specific reports.
- Provide view functionality to allow users to see report details and status.
- Implement a points and badge reward system to encourage user participation.

Project Objectives (solution focus-plan):

These objectives focus on the steps needed to design, develop, and complete the Rasheed system. They describe the main activities required to deliver a functional and effective solution.

- Analyze water and electricity usage issues within the community.
- Gather and define system requirements based on user needs.
- Design simple and user-friendly website interfaces.
- Design and implement the database to store reports and user information.
- Develop the core system features (issue reporting, tracking, and reward system).
- Test the system to ensure proper functionality and fix any detected problems.
- Improve the system based on testing results and user feedback.

Learning Objectives (student focus):

These objectives focus on the skills and knowledge the team will gain throughout the development of the Rasheed project, including technical tools and practical experience.

- Learn how to use project management tools such as Jira to organize tasks and track progress.
- Use GitHub for version control and team collaboration.

- Improve web development skills (HTML, CSS, JavaScript,…).
- Gain experience in designing and implementing databases.
- Practice software development life cycle stages such as analysis, design, implementation, and testing.
- Enhance teamwork and communication skills while working in a Scrum environment.

1.3.4 Scope

In this section, we define the boundaries of the Rasheed system and the functionalities included within the project scope.

Rasheed is a web-based platform designed to serve residents and citizens in Saudi Arabia aged 12 years and above, aiming to simplify the process of reporting water and electricity usage issues. The system allows users to submit reports by describing the problem, providing general location details such as neighborhood and street name, and attaching images when needed. Users can also view, search, edit, and track reported issues through a centralized system. In addition, Rasheed includes a simple reward system where users earn points and badges for verified reports to encourage community participation. The platform supports the English language and is accessible through standard web browsers.

On the other hand, Rasheed does not support map-based location services or GPS tracking. Users are only able to enter general location information manually without the use of interactive maps or automatic location detection.

1.4 The Scrum Team

Table 1: The Scrum Team

Scrum team

Product Owner:	Lama Almubarak
Developers:	Lama Almubarak Tala Alqahtani Shatha bin mana Jood Alkhneen
Scrum Master (SM):	Dr. Reem Alqifari
Stakeholder:	Dr. Reem Alqifari

2 Chapter 2: Domain Analysis

2.1 Literature Review

In this section, existing platforms related to reporting water and electricity issues in Saudi Arabia are reviewed and analyzed to examine their strengths and limitations. The analysis focuses on three main systems: Balady Platform, the National Water Company (NWC) application, and the Saudi Electricity Company (SEC) application. The purpose of this review is to understand the current market solutions, identify their key features, and highlight existing gaps. This evaluation helps justify the need for the proposed Rasheed system and clarifies how it can provide additional value compared to the competing platforms.

Competitor 1:

Balady Platform [3]:

Balady is an official government platform that allows citizens and residents in Saudi Arabia to submit municipal complaints and service requests, including infrastructure and public utility issues.

Strengths:

1. Official government platform.
2. Supports multiple types of complaints.
3. Integrated with municipal authorities.
4. Available as a mobile application.

Limitations:

1. Complex interface for some users.
2. Not specifically focused on water and electricity sustainability.
3. No reward or community engagement system.
4. Limited transparency in issue prioritization.

Competitor 2:

National Water Company (NWC) Application [4]:

The NWC application allows customers to manage water services, submit service requests, and report water-related issues.

Strengths:

1. Specialized in water services.
2. Enables service requests and billing management.
3. Official and trusted source.
4. Allows tracking of requests.

Limitations:

1. Focused only on water.
2. No sustainability awareness features.
3. No community reward system.
4. Limited user engagement features.

Competitor 3:

Saudi Electricity Company (SEC) Application [5]:

The Saudi Electricity Company (SEC) application allows customers to manage electricity services, report outages, and access billing information.

Strengths:

1. Specialized in electricity services.
2. Enables reporting power outages.
3. Allows bill management and service requests.
4. Official and trusted platform.

Limitations:

1. Focused only on electricity.
2. No reward or engagement system.
3. No sustainability awareness features.
4. Limited community-based reporting functionality.

Table 2: Competing software and their features comparing with Rasheed

Feature	Balady Platform	NWC App	SEC App	Rasheed
Report water issues	✓	✓		✓
Report electricity issues	✓		✓	✓
View report details	✓	✓	✓	✓
Delete a report				✓
Search reports	✓			✓
Upload image	✓	✓	✓	✓
Track report status	✓	✓	✓	✓
Select severity level				✓
Reward system				✓

How Rasheed is Different from Competing Systems? :

Although Balady Platform, NWC App, and SEC App provide official channels for reporting utility-related issues, they mainly focus on delivering specific administrative services within separate domains. Balady offers general municipal complaint services, while NWC and SEC focus individually on water and electricity services.

Rasheed, however, integrates both water and electricity reporting within a single platform designed to enhance community engagement and sustainability awareness. In addition to core reporting functionalities, Rasheed introduces features such as severity classification and a reward system that encourage proactive participation and responsible resource usage.

As illustrated in Table 2, existing platforms lack integrated services and user engagement features, which Rasheed addresses by combining unified reporting, user-centered design, and motivational elements.

By combining unified reporting, user-centered design, and motivational elements, Rasheed provides a more engagement-focused and sustainability-driven solution compared to existing systems.

3 Chapter 3: Requirements Engineering

3.1 Requirements Elicitation and Analysis

In this section, the methods used to elicit and analyze the requirements of the Rasheed system are presented, along with the sources of information that informed the identification of functional and non-functional requirements. Requirements elicitation is a critical phase in system development, as it ensures that the defined requirements are grounded in real user needs and existing system limitations. To achieve comprehensive requirement identification, multiple elicitation techniques and information sources were employed.

First, a domain and competitor analysis was conducted to examine existing platforms, including the Balady Platform, the National Water Company (NWC) application, and the Saudi Electricity Company (SEC) application. This analysis provided insights into current reporting mechanisms, identified strengths such as official integration and service specialization, and highlighted limitations including service fragmentation, limited user engagement mechanisms, and the absence of sustainability-driven features.

Second, structured interviews were conducted with selected residents in Saudi Arabia representing diverse age groups and technical backgrounds. These interviews aimed to explore users' experiences with reporting water and electricity issues, the challenges encountered in existing platforms, and their expectations regarding usability, transparency, and feedback mechanisms.

Additionally, a questionnaire was distributed to a broader group of participants to validate recurring themes identified in the interviews and to collect quantitative data regarding reporting behavior, usability preferences, and engagement features.

Relevant documentation, including national sustainability objectives such as in Vision 2030 initiatives, were reviewed to ensure alignment with broader community and environmental goals.

The findings derived from these elicitation activities form the foundation for the system requirements presented in the following sections.

Interviews:

Semi-structured interviews were conducted with four residents in Saudi Arabia representing different age groups, occupations, and levels of technical experience. Each participant was asked a set of six open-ended questions focusing on their experiences with reporting water and electricity issues, the challenges they encountered when using existing platforms, and their expectations for an improved reporting system.

The interview questions were as follows:

1. Have you ever faced a water leak or electricity issue in your area? Did you report it?
2. What difficulties did you face while reporting the issue?
3. Have you ever decided not to report an issue? Why?
4. How important is it for you to track the status of your reported issue?
5. Would a reward system (points or badges) encourage you to report issues more often?
6. What features would make a reporting website easy and user-friendly for you?

The interviews were conducted individually through informal sessions, either in person or via phone communication, to allow participants to provide detailed feedback. The collected responses were analyzed to identify recurring themes and requirement patterns relevant to the Rasheed system.

Participants

Table 3: Summary of Interview Participants

Interviewee ID	Age	Gender	Nationality	Occupation	Technical Experience
P1	46	Male	Saudi	Doctor	Moderate
P2	27	Female	Saudi	Cloud Infrastructure Engineer	High

P3	23	Male	Saudi	Employee	High
P4	45	Female	Saudi	Teacher	Moderate

Findings and Analysis

The interviews were conducted with four participants representing different age groups (23–46 years), professional backgrounds, and varying levels of technical expertise. This diversity allowed for the identification of recurring patterns across users with different lifestyles and digital familiarity.

A dominant theme identified across all interviews was the perceived lack of clarity and efficiency in the reporting process. Participants expressed uncertainty regarding which authority to contact or which platform to use when facing water or electricity issues. In several cases, informal communication methods were used, or issues were left unreported due to the assumption that someone else would handle them. Additionally, users highlighted that existing reporting processes often require repeated data entry and multiple steps, which discourages timely reporting.

Another strongly recurring pattern was the importance of transparency and feedback after report submission. All participants emphasized the need to receive confirmation that their report has been received and track its status through clear progress updates. The absence of tracking mechanisms was viewed as a major limitation that reduces trust in the reporting process. These findings directly support the status tracking and notification features within the Rasheed system to enhance transparency and user trust.

Usability and simplicity were also identified as critical success factors. Participants across all age groups preferred a clean interface with minimal typing, saved user information, quick access buttons, and mobile-friendly functionality. Even users with high technical proficiency expressed frustration with unnecessarily complicated processes. Consequently, usability considerations such as simplified navigation and reduced form complexity were prioritized in the system design.

Behavioral insights also emerged from the interviews. Several participants admitted that they sometimes avoid reporting minor issues due to time constraints or the belief that others will report them. This suggests that reducing reporting friction could increase user participation.

Regarding the reward system, participants generally indicated that incentives such as points or badges are not the primary motivation for reporting issues. However, they acknowledged that rewards could enhance engagement, encourage faster reporting, and create a sense of appreciation and recognition.

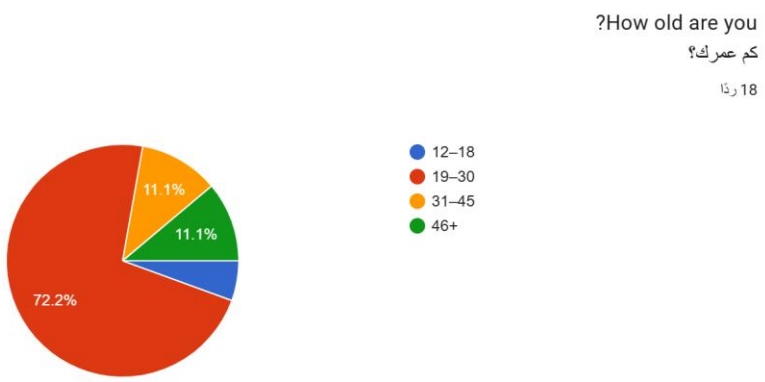
Overall, the findings confirm the need for a centralized, transparent, and user-friendly reporting platform that minimizes effort, provides real-time tracking, and encourages community participation through supportive engagement mechanisms. These insights provided qualitative validation for the proposed functional and non-functional requirements outlined in the subsequent sections.

The detailed interview responses and full transcriptions are provided in Appendix A.

Questionnaire:

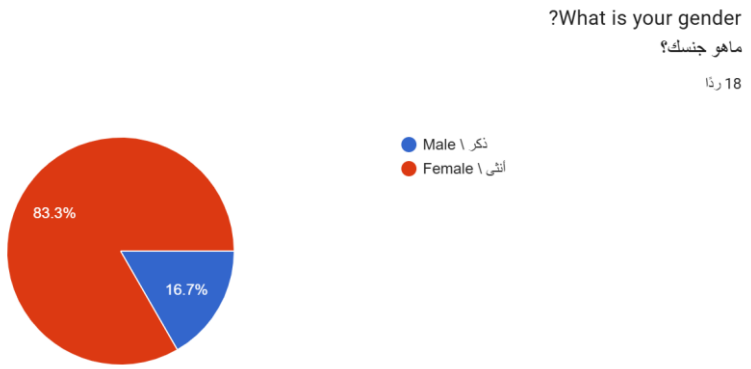
The questionnaire was conducted to collect quantitative data regarding user experiences and expectations related to reporting water and electricity issues. A total of 18 participants responded to the questionnaire, exceeding the minimum requirement of 10 participants. The questionnaire consisted of more than five structured questions designed to gather insights about reporting behavior, usability preferences, and potential adoption of a web-based platform.

The results are presented below in graphical format, followed by analysis for each question.



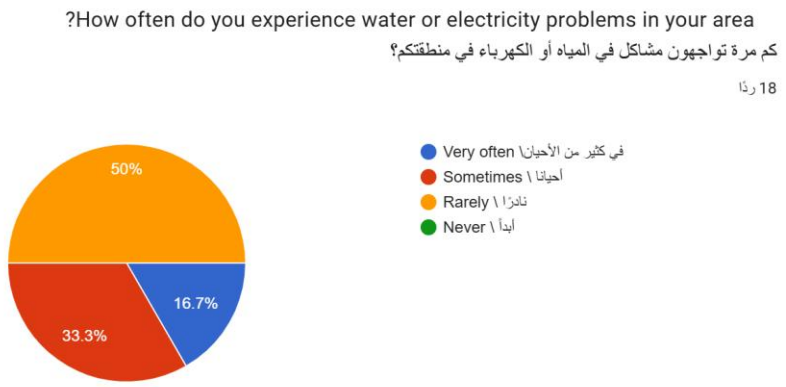
Age Distribution

The results show that the majority of participants (72.2%) fall within the 19–30 age group, indicating that young adults were the most represented group in the questionnaire. However, Rasheed is designed to serve users aged 12 years and above, which requires the platform to maintain a simple and accessible interface suitable for teenagers as well as older users. The diversity in age groups highlights the importance of usability and clarity in system design.



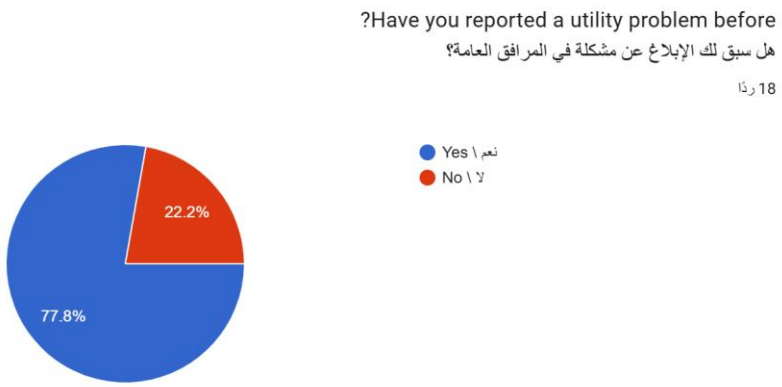
Gender Distribution

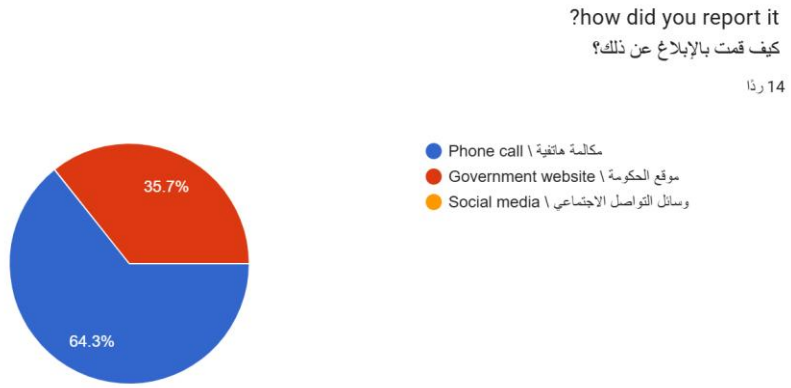
The data shows that 83.3% of respondents were female. This may suggest that women are more actively involved in handling household-related utility issues such as water leaks or electricity interruptions. Therefore, the system interface should be intuitive and straightforward to support quick reporting in daily-life contexts



Frequency of Utility Problems

Although 50% of respondents stated that such issues occur rarely, 33.3% experience them sometimes, and 16.7% very often. This indicates that while the problems are not constant, they are recurring enough to justify the need for a structured and efficient reporting platform.





Reporting Behavior

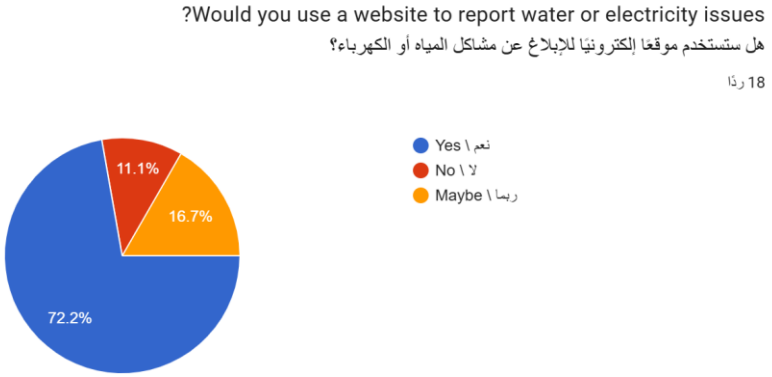
A significant 77.8% of participants have previously reported utility issues. This demonstrates user willingness to engage with public service systems. However, 64.3% relied on phone calls as the primary reporting method, indicating that current processes may lack digital accessibility or convenience. This finding supports the need for a centralized web-based alternative such as Rasheed.

Ease of Reporting

While most participants described the reporting process as easy (64.3%), a noticeable percentage (14.3%) found it difficult. This suggests inconsistencies in the user experience across existing channels, emphasizing the importance of usability-focused design in Rasheed.

Resolution Time

Half of the participants indicated that issues are resolved within a few days, while 21.4% believe the resolution takes a long time. This reflects a need for enhanced transparency and tracking mechanisms within the reporting process.



Website Adoption

72.2% expressed willingness to use a website for reporting issues. This provides strong validation for the Rasheed platform concept and indicates potential acceptance among users.

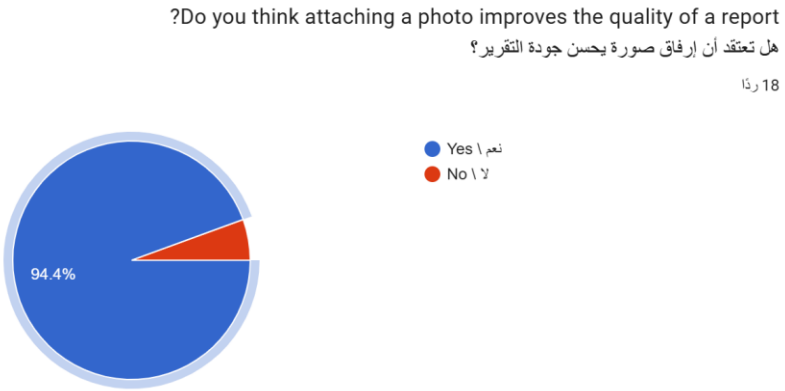
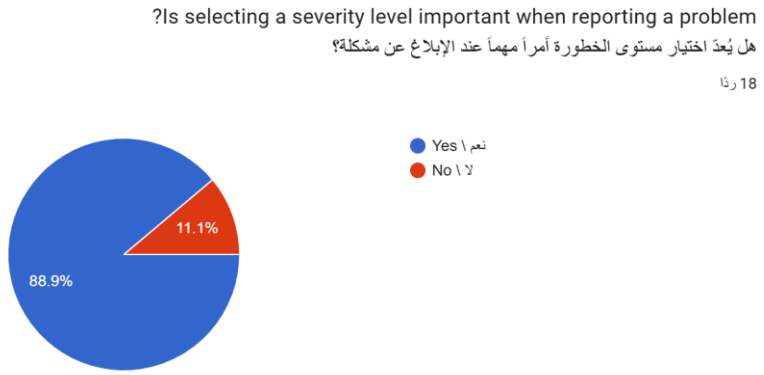


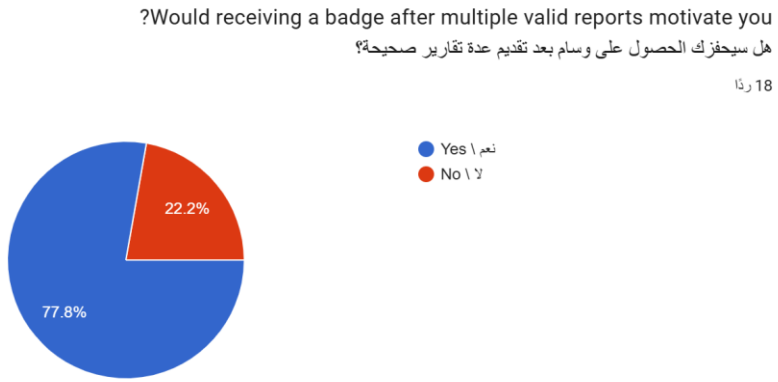
Photo Attachment Importance

An overwhelming 94.4% agreed that attaching a photo improves report quality. This highlights the critical importance of integrating an image-upload feature to enhance clarity and speed up problem resolution.



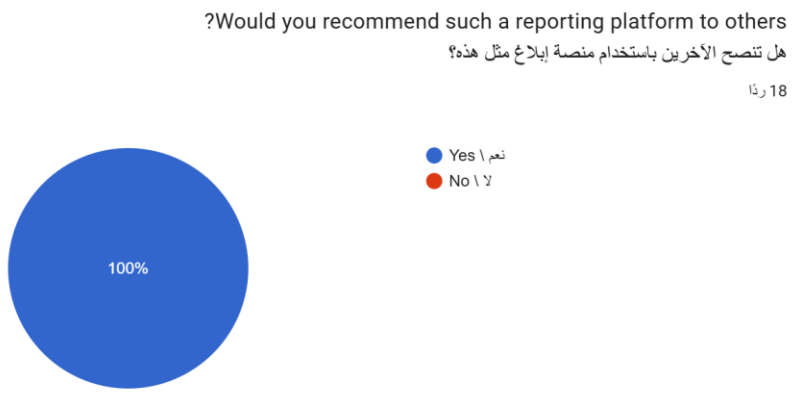
Severity Level Selection

88.9% consider selecting a severity level important when reporting a problem. This supports implementing a priority classification system (e.g., low, medium, high) to facilitate efficient handling and resource allocation.



Motivation Through Badges

77.8% stated they would be motivated by receiving badges after submitting valid reports. This suggests that incorporating gamification elements may increase user engagement and encourage continued participation.



Recommendation Intention

100% of participants stated they would recommend such a platform. This reflects strong perceived value and potential community acceptance of Rasheed.

Summary of Findings

The questionnaire results indicate strong support for a simple, centralized, and transparent digital reporting system. Participants emphasized the importance of ease of use, photo attachment, severity classification, and status tracking. Additionally, the high willingness to adopt and recommend the platform confirms the feasibility of Rasheed as a web-based solution. These findings directly informed the functional requirements of the system, including online report submission, image upload capability, severity level selection, report tracking, and gamification features.

3.2 System Users

Based on the findings from requirements elicitation and analysis, the key system users were identified and categorized according to their roles and responsibilities within the Rasheed system. This section provides a description of each user type, including their roles, goals, and technical background. Defining system users helps ensure that the platform is designed to meet the needs of all stakeholders effectively.

1. User (General User)

The User represents any registered individual who interacts with the Rasheed system. This role captures shared authentication-related functionalities.

The following characteristics apply to all users of the Rasheed system:

- Age: Users are 12 years old and above to ensure they are capable of independently interacting with the platform.
- Nationality: The system primarily targets residents within Saudi Arabia, as it addresses local water and electricity issues.
- Educational Level: Users are expected to have at least a basic educational background that enables them to understand instructions and interact with digital forms.
- Technical Expertise: Users are expected to possess basic digital literacy skills, such as using smartphones, web browsers, and navigating simple user interfaces.
- Language Proficiency: Basic proficiency in English is required, as the current system interface is available in English.

Goals:

- Access the system securely.
- Navigate platform features efficiently.
- Stay informed about system updates and notifications.

2. Resident (System/Specialized User)

A Resident is a specialized type of User who actively submits and manages reports related to water and electricity issues within their community.

Domain Experience: Residents may or may not have prior experience in reporting utility-related issues but are familiar with encountering such problems in daily life.

Goals:

- Report utility issues quickly and easily.
- Monitor the progress of submitted reports.
- Contribute to sustainability efforts within the community.

- Receive recognition through the reward mechanism.

3. Admin (Administrator)

An Admin is an authorized user responsible for managing and processing submitted reports within the Rasheed system.

Domain Experience: Administrators are expected to be familiar with report validation processes and utility issue management procedures.

Technical Requirements: Administrators require moderate technical proficiency to operate management dashboards and update report statuses accurately.

Goals:

- Ensure efficient and transparent report processing.
- Maintain fairness and accuracy in status updates.
- Improve response efficiency and operational effectiveness.

3.3 Use Case Diagram

This section presents the Use Case Diagram of the Rasheed System, illustrating the main functionalities of the system and the interactions between its actors (User, Resident, and Admin). The diagram provides a clear overview of how each actor interacts with the system's core features.

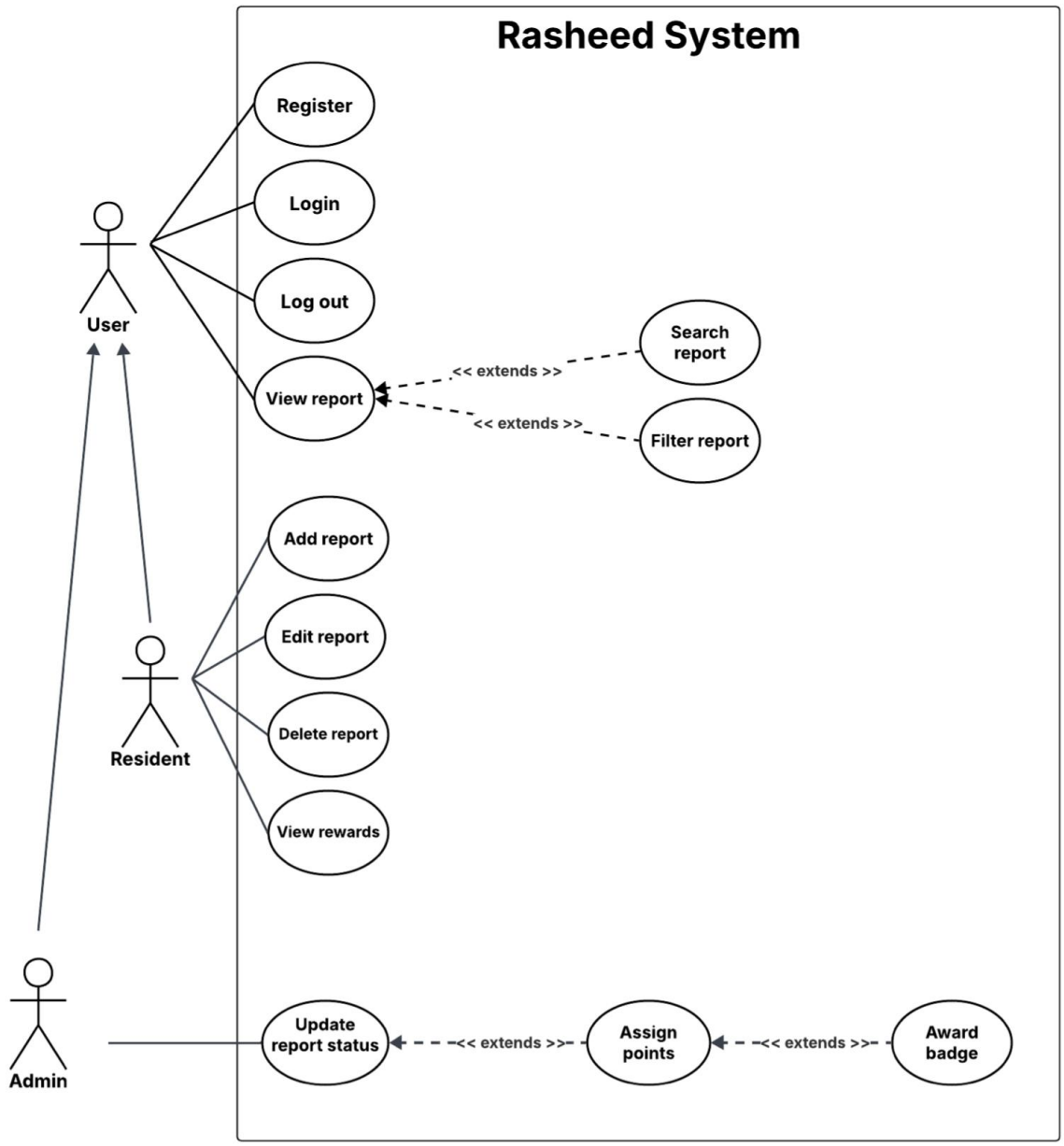


Figure 2: Rasheed Use Case diagram

3.4 Product Backlog

This section presents the product backlog of Rasheed Platform, which includes the system features written in user story format.

Each Product Backlog Item (PBI) includes its size (story points), type, and acceptance criteria.

The backlog corresponds to the items created in the Jira project.

Table 4 :Rasheed Product backlog

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
As a guest user, I want to register for an account so that I can submit and track my reports	3	Feature	• The registration form should include (Full name, Phone number, Password (minimum 8 characters)) • The system should validate the phone number format • The password must include at least (One uppercase letter, One number) • The system should prevent duplicate phone numbers • After successful registration, the user should be redirected to the home page
As a registered user, I want to log in with my phone number and password so that the system can authenticate me and allow me to access my reports	2	Feature	• If I enter correct credentials and click Log in, then I should be redirected to my dashboard. • If I enter incorrect credentials and click Log in, then the system should show a general error message. • After logging in successfully, I should be able to view my reports.
As a registered user, I want to log out so that I can end my session	1	Feature	• If I click on the Log out button, then the system should log me out successfully.

			• If I try to open a protected page after logging out, then I should be redirected to the login page.
As a resident, I want to delete my report before it is completed so that I can remove incorrect or unnecessary submissions	3	Feature	• The system shall allow the resident to delete a report only if its status is not "Completed". • Once deleted, the report shall no longer appear in the resident's report list. • The system shall display a confirmation message before deleting the report. • If the report status is "Completed", the delete option shall not be available. • Deleted reports shall remain stored in the system database for administrative purposes.
As a resident, I want to add a report so that the admin can review and take action	5	Feature	• The resident shall be able to select the issue type (Water / Electricity). • The resident shall be able to enter a description of the issue. • The resident shall be able to enter location details (Neighborhood, Street name, Building number). • The resident shall be able to upload an image (optional).

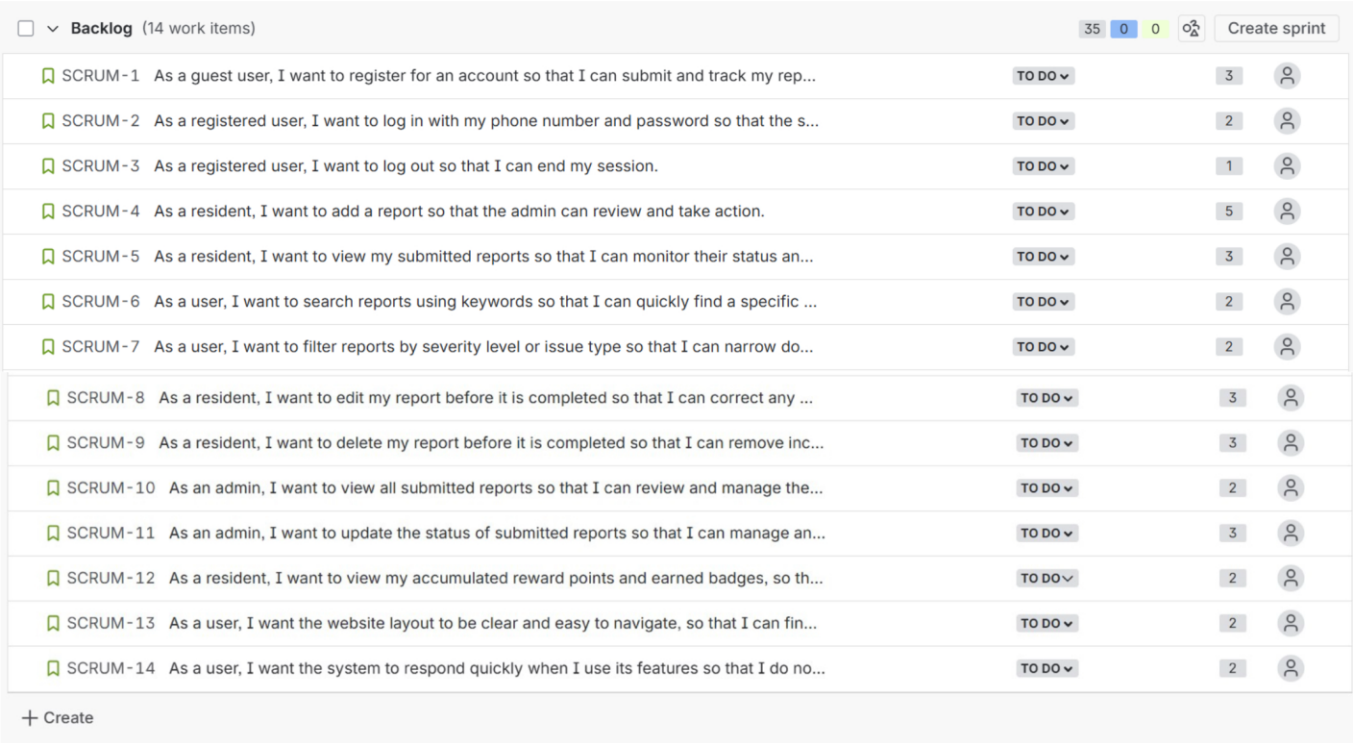
			• The resident shall be able to select a severity level (Low, Medium, High). • After successful submission, the system shall display a confirmation message. • The submitted report shall be stored in the system database.
As a resident, I want to view my submitted reports so that I can monitor their status and track their progress.	3	Feature	• The system shall display a list of reports submitted by the resident. • Each report shall display its current status (Pending / In Progress / Completed). • The resident shall be able to open a report to view detailed information. • The displayed report status shall be updated when changed by the admin.
As a resident, I want to edit my report before it is completed so that I can correct any mistakes.	3	Feature	• The system shall allow the resident to edit a report only if its status is not “Completed”. • The resident shall be able to update the report description, location details, severity level, and image (optional). • The system shall save the updated report information in the database. • The system shall display a confirmation message after saving changes.

			• If the report status is “Completed”, the edit option shall not be available.
As a user, I want to search reports using keywords so that I can quickly find a specific report.	2	Feature	• The system shall allow the user to search reports by keyword. • The system shall display reports that match the search input. • If no results are found, the system shall display a “No reports found” message.
As a user, I want to filter reports by severity level or issue type so that I can narrow down the results.	2	Feature	• The system shall allow filtering by severity level. • The system shall allow filtering by issue type. • The system shall display only reports matching selected criteria. • The user shall be able to clear filters.
As a resident, I want to view my accumulated reward points and earned badges, So that I feel motivated to continue reporting issues.	2	Feature	• The system shall display the resident’s total points. • The system shall display earned badges. • Points shall be updated only after admin changes report status to “Completed”.
As an admin, I want to view all submitted	2	Feature	• The system shall display a list of all submitted reports.

reports so that I can review and manage them.			• The admin shall be able to open any report to view its details. • The system shall allow the admin to view report status and submission information.
As an admin, I want to update the status of submitted reports so that I can manage and process them properly	3	Feature	• The admin shall be able to change the report status to (Pending / In Progress / Completed / Deleted). • The updated status should be visible to the report owner. • The system shall save the updated status in the database. • When the status changes, the system shall generate a notification for the report owner. • When the status is changed to "Completed," the system shall award reward points to the report owner (only once per report).
As a user, I want the website layout to be clear and easy to navigate, So that I can find features quickly without confusion.	2	Feature (Non-functional–Usability)	• The main navigation menu shall include clearly labeled buttons (Submit Report, My Reports, Notifications, Rewards, Logout). • The user shall be able to access major features within 2-3 clicks. • The interface shall not contain unnecessary or duplicate options.
As a user, I want the system to respond quickly when I use its	2	Feature (Non-functional–Performance)	• The system shall load the user dashboard within 3 seconds under normal operating conditions.

features so that I do not experience delays.			• The system shall display the list of reports within 3 seconds. • The system shall process report submission within 5 seconds.

Figure 3:Rasheed Product backlog in jira



4 Chapter 4: System Design and Development

4.1 System Architecture

The Rasheed system is designed using a Client-Server architecture with a layered structure to support a centralized and efficient reporting platform for water and electricity issues. This architecture enables clear separation between user interaction, system processing, and data management, which is essential for handling report submissions, status tracking, and user engagement features such as notifications and rewards.

The system is organized into three main layers: the Presentation Layer, the Business Logic Layer, and the Data Layer, where each layer is responsible for a specific set of functionalities within the system.

Presentation Layer (Client Side):

The Presentation Layer represents the user interface through which users interact with the system. It is designed to be simple and user-friendly to accommodate users with different levels of technical experience. Through this layer, users can:

- Register and log in to the system
- Submit reports related to water or electricity issues
- View and track report status
- Receive notifications
- View points and badges

This layer captures user input and sends requests to the system.

Business Logic Layer (Server Side):

The Business Logic Layer is responsible for processing all system operations and enforcing system rules. It handles operations initiated by users, such as:

- Validating user input
- Processing report submissions by residents
- Processing report status updates initiated by the admin
- Managing rewards and point calculations

- Generating notifications

The server processes user requests and ensures that all system operations are executed correctly.

Data Layer:

The Data Layer is responsible for managing data storage and retrieval. It includes:

- User data
- Reports
- Notifications
- Rewards and badges

This layer ensures data consistency, integrity, and efficient access.

User actions, such as submitting a report or updating its status, are initiated through the Presentation Layer. These actions are processed in the Business Logic Layer, where system rules are applied, and then stored or retrieved from the Data Layer. The processed results are then returned to the user interface.

In conclusion, the layered Client-Server architecture provides a clear separation between system components, ensuring scalability, maintainability, and reliable system performance for the Rasheed system.

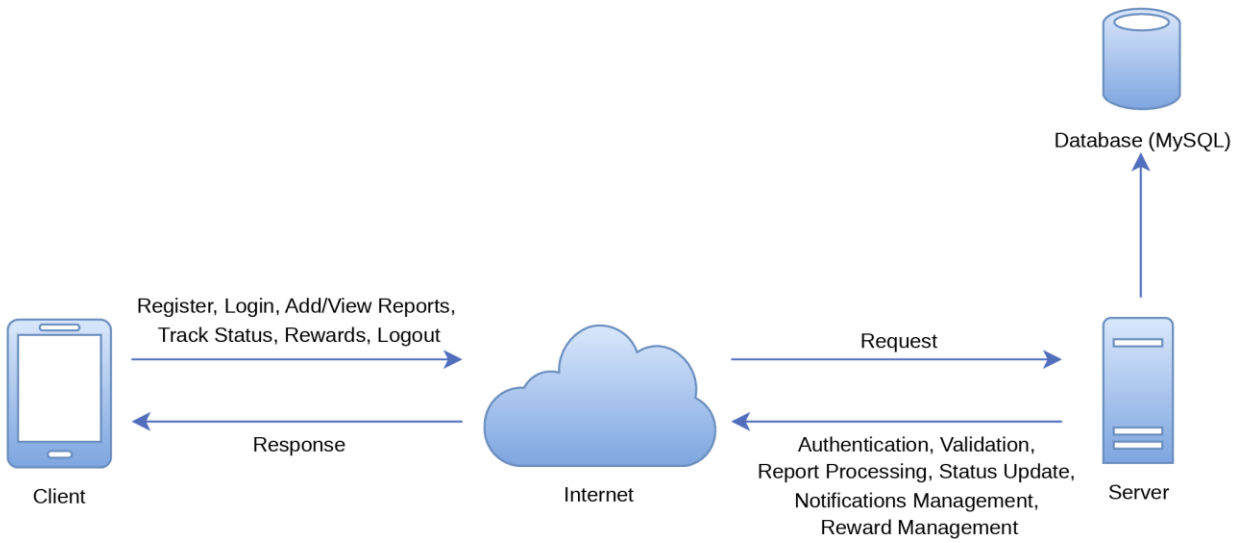


Figure 4: System Architecture of Rasheed

4.2 Class Diagram

Figure 5 illustrates the complete class diagram of the Rasheed system, highlighting modules for user management, report handling, and the reward system. It shows how entities such as User, Resident, Admin, Report, and Badge interact to support authentication, report submission, validation, and reward allocation, ensuring efficient data flow and system scalability.

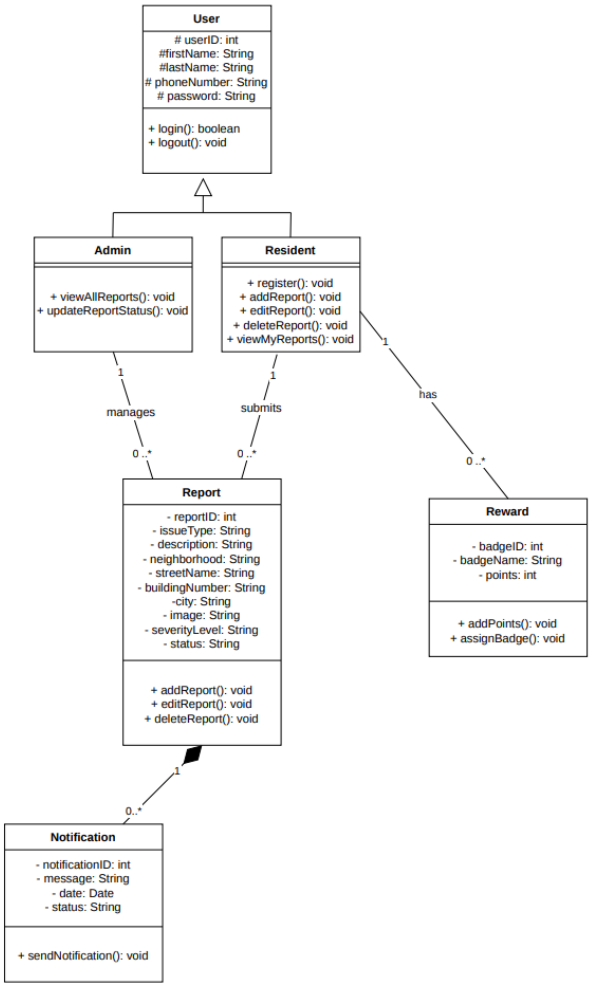


Figure 5:Rasheed Application Class Diagram

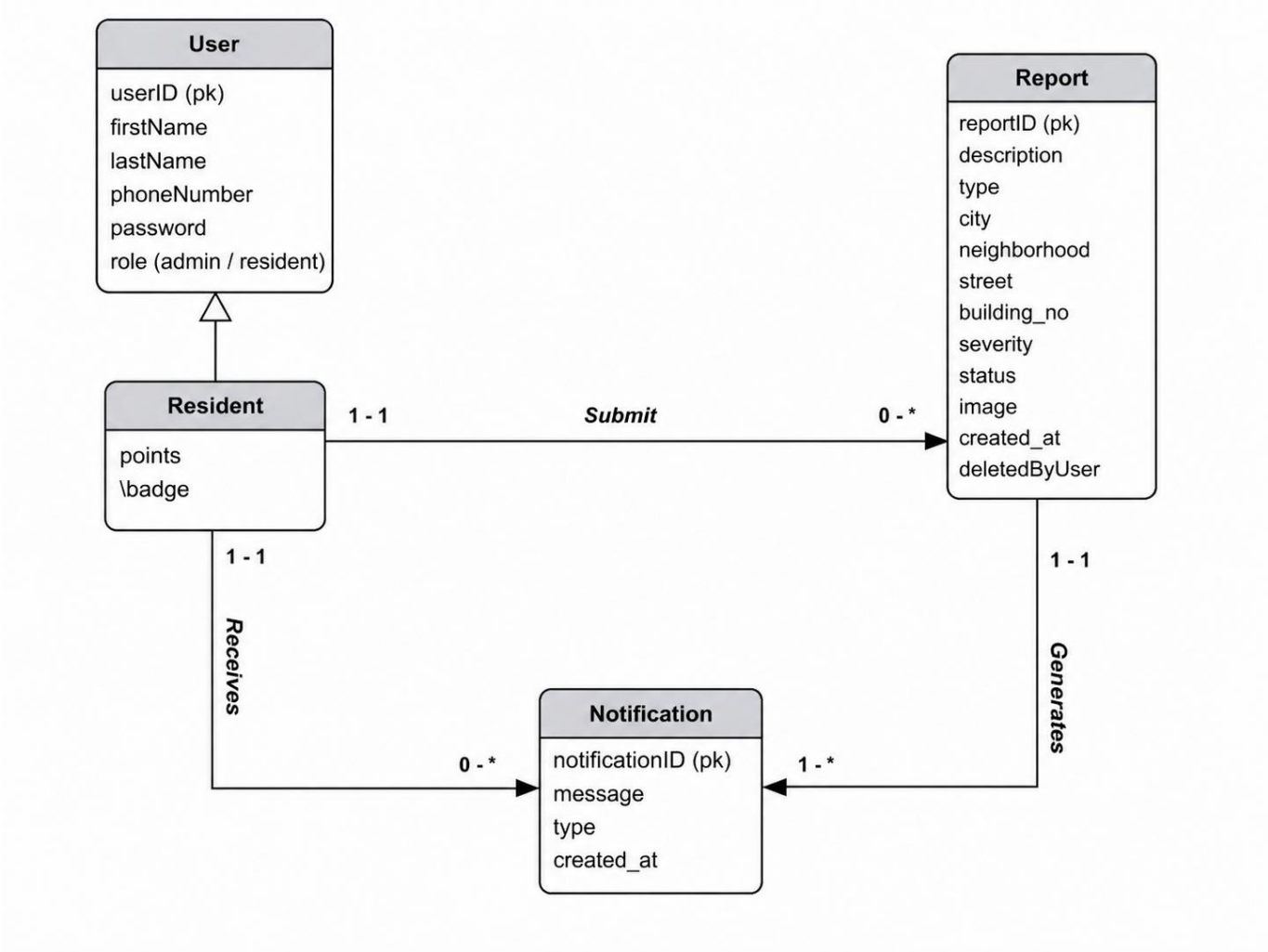
4.3 Data Design

In this section, we present the data design of the Rasheed system, which defines how data is structured and managed to support key system functionalities such as report submission, status tracking, notifications, and rewards.

This section includes the ER Diagram, the Schema, and the Data Dictionary, which together describe the system's entities, relationships, and data structure. These components ensure data consistency, integrity, and efficient access within the system.

- **ER Diagram**

Figure 6: ER diagram of Rasheed



- **Schema**

User (userID, firstName, lastName, phoneNumber, password, role)

Primary Key: userID

Resident (residentID, points)

Primary Key: residentID

Foreign Key: residentID references User(userID)

Report (reportID, description, type, city, neighborhood, street, building_no, severity, status, image, created_at, deletedByUser, residentID)

Primary Key: reportID

Foreign Key: residentID references Resident(residentID)

Notifications (notificationID, message, type, created_at, residentID, reportID)

Primary Key: notificationID

Foreign Key: residentID references Resident(residentID)

Foreign Key: reportID references Report(reportID)

- **Data Dictionary**
  - Data Dictionary showing description of all entities

Table 5: Data Dictionary showing description of all entities

Entity Name	Description	Occurrence
User	Represents a system user who can be either a resident or admin.	A user can be a resident and interact with the system.
Resident	Represents a user who submits reports and receives notifications.	A resident can submit multiple reports and receive multiple notifications.
Report	Represents an issue submitted by a resident (water/electricity).	A report belongs to one resident.
Notification	Represents a message generated based on report updates.	A notification is generated for reports and received by residents.

  - Data Dictionary showing description of all relationship

Table 6: : Data Dictionary showing description of all relationships

Entity	Multiplicity	Relationship	Entity	Multiplicity
User	1..1	Is a	Resident	1..1
Resident	1..1	submits	Report	0..*
Resident	1..1	receives	Notification	0..*
Report	1..1	generates	Notification	1..*

– Data Dictionary showing description of all attributes

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Valued	Default Value	Range	PK
User	userID	User's ID	String	255	N	N	N		Y
	firstName	User's first name	String	50	N	N	N		
	lastName	User's last name	String	50	N	N	N		
	phoneNumber	User's phone number	String	15	N	N	N		
	password	User's password	String	100	N	N	N	6-100	
	role	Admin / Resident	String	20	N	N	N		
Report	reportID	Report ID	String	255	N	N	N		Y
	description	Issue description	String	255	N	N	N		
	type	Type of issue	String	50	N	N	N		
	city	City	String	50	N	N	N		
	neighborhood	Neighborhood	String	50	N	N	N		
	street	Street	String	50	N	N	N		
	buliding_no	Building number	String	20	N	N	N		
	severity	Issue severity	String	20	N	N	N		
	status	Report status	String	20	N	N	N		
	image	Report image	String	255	Y	N	N		
	created_at	Created date	DateTime	-	N	N	N		
	deletedByUser	Indicates if report deleted by resident or admin	Boolean	-	N	N	0	0 or 1	
	residentID (FK)	Reference to Resident	String	255	N	N	N		

Resident	residentID (FK)	Reference to Resident	String	255	N	N	N		Y
	points	Points earned	Integer	50	N	N	N	0-1000	
	badge	User badge or level	String	50	N	N	N		
Notifications	notificationID	Notifications ID	String	255	N	N	N		Y
	message	Notifications message	String	50	N	N	N		
	type	Notifications type	String	50	N	N	N		
	created_at	Created date	DateTime	-	N	N	N		
	residentID (FK)	Reference to Resident	String	255	N	N	N		
	ReportID (FK)	Reference to Report	String	255	N	N	N		

Table 7: Data Dictionary showing description of all attributes

4.4 Component Design

This section presents the design of key system components. Each component is described using a flowchart and corresponding pseudocode to illustrate how the functionality is implemented. The selected components represent the core operations of the system beyond basic functionalities such as login and registration.

Component 1: **Add a Report**

This component allows the resident to submit a new report related to a water or electricity issue. The process includes entering the required report details, validating the input fields, creating the report, and saving it with an initial status of pending.

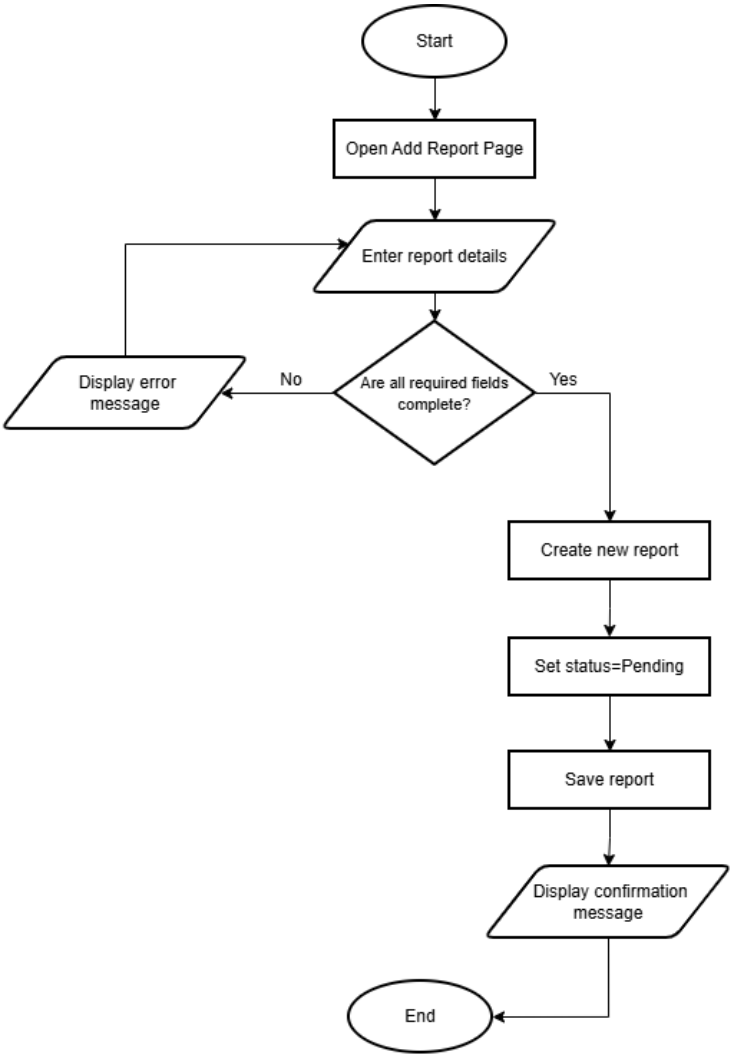


Figure 7: Flowchart for Add Report

Pseudocode:

BEGIN

OPEN Add Report Page

READ issueType, description, neighborhood, streetName, buildingNumber, severityLevel, image

IF issueType is empty OR description is empty OR neighborhood is empty OR streetName is empty OR city is empty OR buildingNumber is empty OR severityLevel is empty

  WRITE "Please complete all required fields"

ELSE

  CREATE report

  SET reportStatus = "Pending"

```
    SAVE report
    WRITE "Report submitted successfully"
ENDIF

END
```

Component 2: **Update Report Status (Admin)**

This component allows the administrator to manage submitted reports by updating their status. After a report is selected, its details are displayed, and the administrator can assign a new status. If the status is updated to completed, the system automatically awards 10 reward points to the report owner.

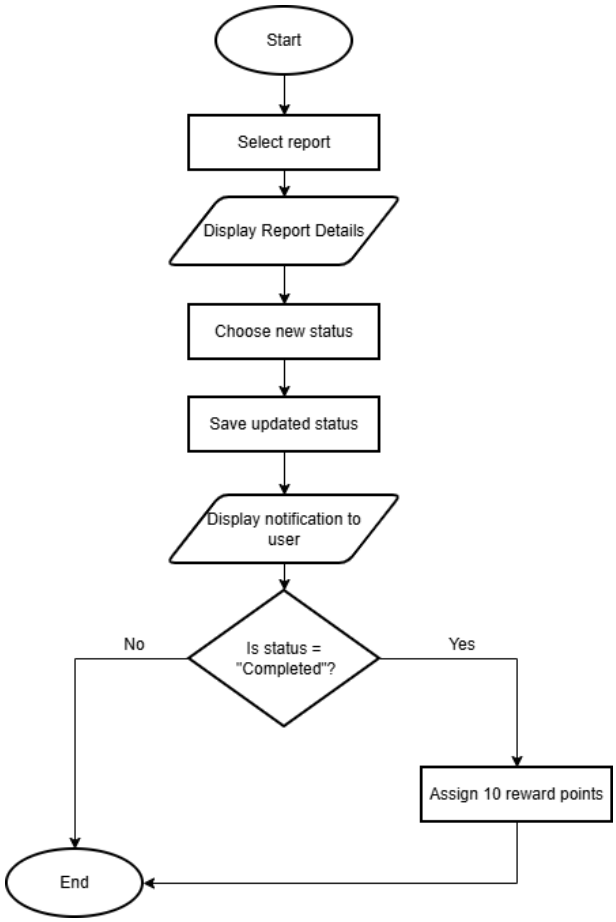


Figure 8: Flowchart for Updating Report Status

Pseudocode:

```
BEGIN
```

```
READ selectedReport
WRITE report details
READ newStatus

SAVE updatedStatus
WRITE "Notification sent to the report owner"

IF newStatus = "Completed"
   ADD 10 points to reportOwner
ENDIF

END
```

Component 3: **Track Report Status**

This component allows the resident to track the status of previously submitted reports. The resident opens the reports page, selects a report, and the system displays its details along with the current status.

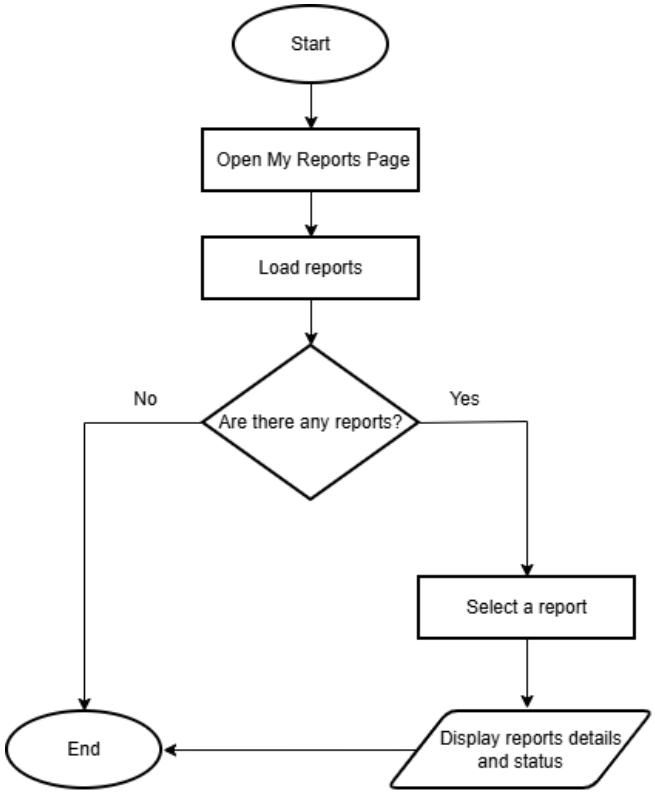


Figure 9:Flowchart for Tracking Report Status

Pseudocode:

```
BEGIN

OPEN My Reports Page
LOAD submitted reports

IF reports are available
   READ selectedReport
   WRITE report details and current status
ENDIF

END
```

4.5 Interface Design

This section presents the interface design of the Rasheed system. A navigation diagram is used to illustrate the structure of the application and how users navigate between different pages. The system consists of separate interfaces for residents and administrators, each providing access to specific functionalities.

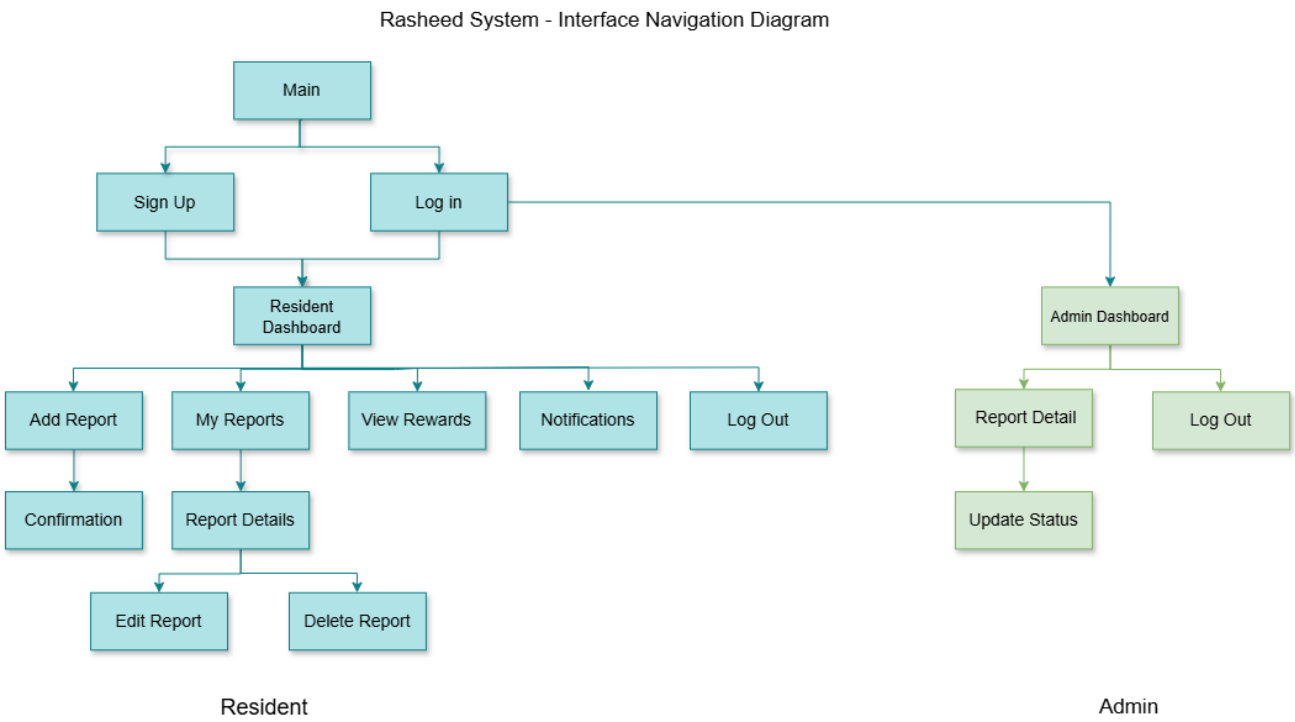


Figure 10: Navigation Diagram of the Rasheed System

As shown in Figure 10, users can access the system through the main page and navigate to login or registration pages. After authentication, residents can access features such as submitting reports, viewing reports, tracking rewards, and receiving notifications. Administrators, on the other hand, can view all reports and update their status.

4.6 Implementation

In this section, the Rasheed system is implemented based on a client-server architecture. Currently, the front-end has been developed to allow users to submit reports and track their status. The back-end and database components, including MySQL, are planned for future implementation to handle request processing, data validation, and report management.

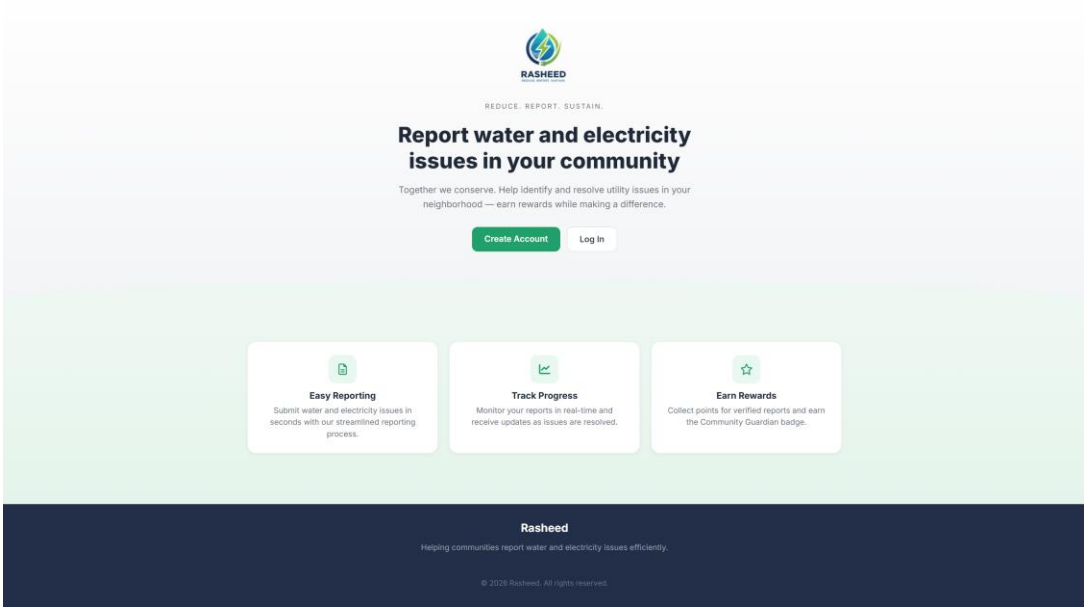


Figure 11:Rasheed Home Page

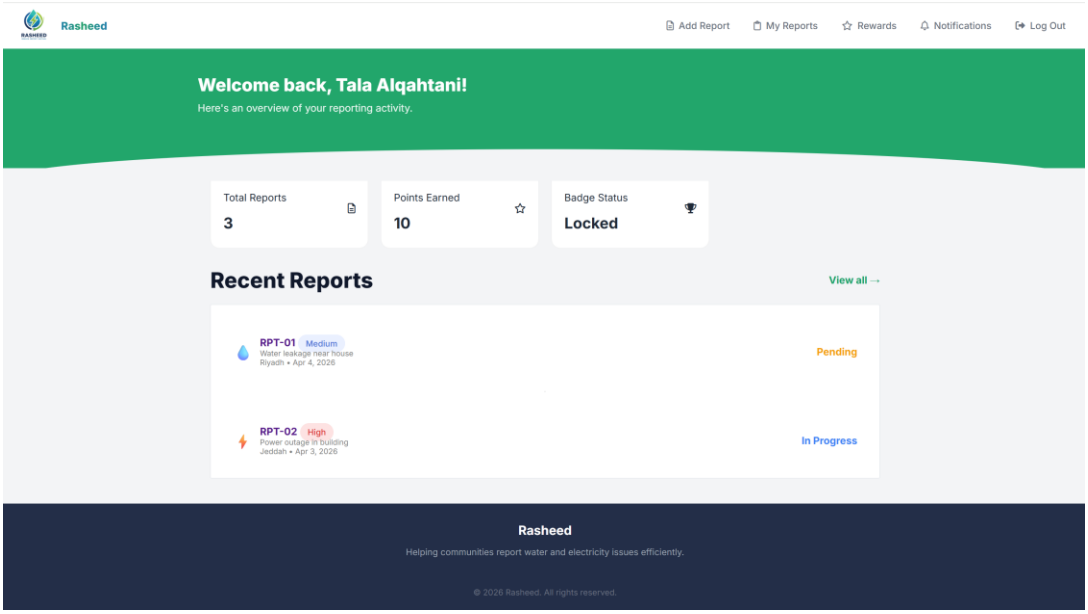


Figure 12: User Dashboard

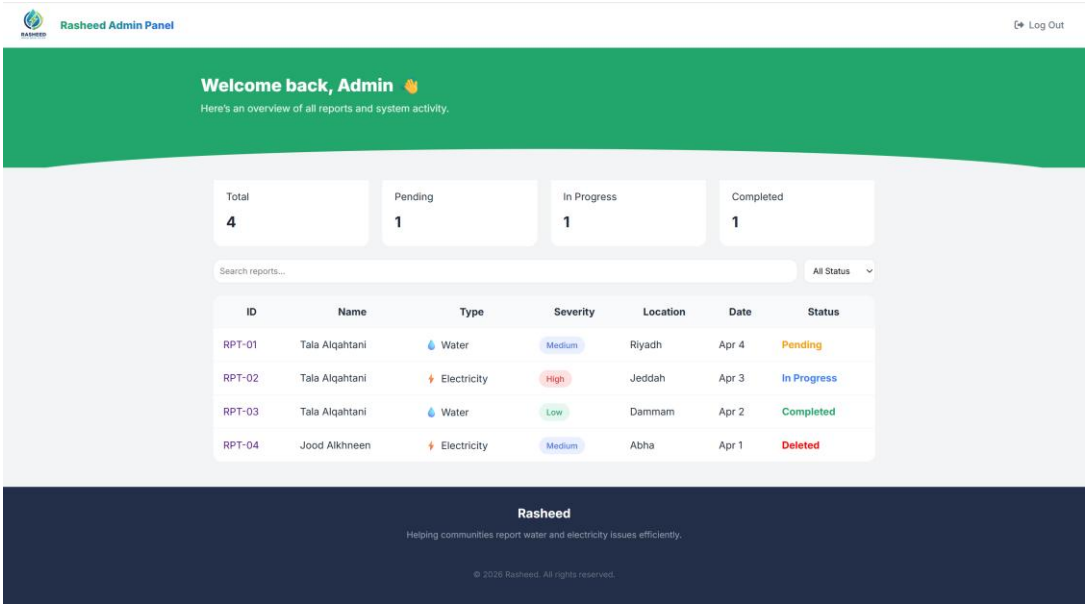


Figure 13: Admin Dashboard

GitHub repository link: https://github.com/talaA2/IT320-Rasheed-System/tree/main

Jira project link: https://it320-project.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog

5 Chapter 5: System Evaluation

5.1 User Story Acceptance Testing

This section presents the User Story Acceptance Testing for the system. The goal is to ensure that each user story meets its acceptance criteria and works as expected. The test cases are based on real user actions, and the results are shown in the table below, indicating whether each user story has passed or not.

Table 8: User Story Acceptance Testing

Sprint Number	User story	Acceptance criteria	Test cases(s)	Pass?	Comments
2	As a guest user, I want to register for an account so that I can submit and track my reports	• The registration form should include (Full name, Phone number, Password (minimum 8 characters)) • The system should validate the phone number format • The password must include at least (One uppercase letter, One number) • The system should prevent	1. User enters valid full name, phone number, and password. 2. User submits the registration form. 3. System creates a new account successfully. 4. Verify that the user is redirected to the home page. 5. User enters an already existing phone number. 6. Verify that the system shows an error message.	Yes	

		duplicate phone numbers • After successful registration, the user should be redirected to the home page			
2	As a registered user, I want to log in with my phone number and password so that the system can authenticate me and allow me to access my reports	• If I enter correct credentials and click Log in, then I should be redirected to my dashboard. • If I enter incorrect credentials and click Log in, then the system should show an error message. • After logging in successfully, I should be able to view my reports.	1. User enters correct phone number and password. 2. User clicks "Log In". 3. System verifies the credentials. 4. Verify that the user is redirected to the dashboard. 5. User enters incorrect password. 6. Verify that the system shows an error message. 7. User enters a phone number that does not exist. 8. Verify that the system shows "User not found".	Yes	

2	As a registered user, I want to log out so that I can end my session	• If I click on the Log out button, then the system should log me out successfully. • If I try to open a protected page after logging out, then I should be redirected to the login page.	1. User clicks on the "Logout" button. 2. Verify that the user is logged out and redirected to login page. 3. User tries to access a protected page after logout and is redirected to the login page.	Yes	
2	As a resident, I want to delete my report before it is completed so that I can remove incorrect or unnecessary submissions	• The system shall allow the resident to delete a report only if its status is not "Completed". • Once deleted, the report shall no longer appear in the resident's report list. • The system shall display a	1. Resident selects a report from "My Reports". 2. Resident clicks "Delete". 3. Resident confirms deletion. 4. System hides the report from the resident's list. 5. Verify that the report still exists in the database.	Yes	

		confirmation message before deleting the report. • If the report status is "Completed", the delete option shall not be available. • Deleted reports shall remain stored in the system database for administrative purposes.			
2	As a resident, I want to add a report so that the admin can review and take action	• The resident shall be able to select the issue type (Water / Electricity). • The resident shall be able to enter a description of the issue. • The resident shall be able to enter location details (Neighborhood,	1. Resident navigates to "Add Report". 2. Resident selects issue type, enters description, location details, and severity level. 3. Resident optionally uploads an image. 4. Resident submits the report. 5. Verify that the report is stored in	Yes	

		Street name, Building number). • The resident shall be able to upload an image (optional). • The resident shall be able to select a severity level (Low, Medium, High). • After successful submission, the system shall display a confirmation message. • The submitted report shall be stored in the system database.	the database and appears in “My Reports”.		
2	As a resident, I want to view my	• The system shall display a list of reports	1. Resident navigates to “My Reports” page.	Yes	

	submitted reports so that I can monitor their status and track their progress.	submitted by the resident. • Each report shall display its current status (Pending / In Progress / Completed). • The resident shall be able to open a report to view detailed information. • The displayed report status shall be updated when changed by the admin.	2. System displays all reports belonging to the resident. 3. Verify that each report displays its correct status (Pending, In Progress, Completed, Deleted). 4. Verify that reports deleted by the resident are not displayed. 5. Verify that reports deleted by the admin are displayed with status “Deleted”.		
2	As a resident, I want to edit my report before it is completed so that I can correct any mistakes.	• The system shall allow the resident to edit a report only if its status is not “Completed”. • The resident shall be able to update the report description, location	1. Resident opens a report with status “Pending”. 2. Resident clicks “Edit”. 3. Resident get confirmation message and updates the report successfully.	Yes	

		details, severity level, and image (optional). • The system shall save the updated report information in the database. • The system shall display a confirmation message after saving changes. • If the report status is "Completed", the edit option shall not be available.	4. Resident opens a "Completed" report. 5. Verify that edit is not allowed. 6. Resident opens a "deleted" report that was deleted by admin. 7. Verify that edit is not allowed.		
2	As a user, I want to search reports using keywords so that I can quickly find	• The system shall allow the user to search reports by keyword. • The system shall display reports that	1. Resident navigates to "My Reports" page. 2. Enters a keyword in the search field. 3. System displays matching reports. 4. If keyword does not match any	Yes	

	a specific report.	match the search input. • If no results are found, the system shall display a “No reports found” message.	report, system shows "No reports found".		
2	As a user, I want to filter reports by severity level or issue type so that I can narrow down the results.	• The system shall allow filtering by severity level. • The system shall allow filtering by issue type. • The system shall display only reports matching selected criteria. • The user shall be able to clear filters.	1. Resident navigates to “My Reports” page. 2. Selects a severity or type filter. 3. System displays filtered reports. 4. Resident clears filters by select “All Types” and “All Severity”. 5. System displays all reports again.	Yes	
2	As a resident, I want to view my accumulated reward points and	• The system shall display the resident’s total points.	1. Resident navigates to the “Rewards” page. 2. The system displays the total accumulated points.	Yes	

	earned badges, So that I feel motivated to continue reporting issues.	• The system shall display earned badges. • Points shall be updated only after admin changes report status to "Completed".	3. Admin updates a report status to "Completed". 4. Resident refreshes the Rewards page and verifies that points are updated. 5. Verify that the badge is unlocked when points reach 100.		
2	As an admin, I want to view all submitted reports so that I can review and manage them.	• The system shall display a list of all submitted reports. • The admin shall be able to open any report to view its details. • The system shall allow the admin to view report status and submission information.	1. Log in as admin and open the Admin Dashboard. Verify that all submitted reports are displayed. 2. Click on a report ID and verify that the report details page opens with full information.	Yes	Reports were displayed successfully, and details were accessible

2	As an admin, I want to update the status of submitted reports so that I can manage and process them properly	• The admin shall be able to change the report status to (Pending/ In Progress/ Completed/ Deleted). • The updated status should be visible to the report owner. • The system shall save the updated status in the database. • When the status changes, the system shall generate a notification for the report owner. When the status is changed to "Completed," the system shall award	1. Open a report as admin, change the status to In Progress, and verify that the new status is saved and displayed correctly. 2. Open a report as admin, change the status to Completed, and verify that: • The status is updated. • The resident receives a notification. • 10 points are added. 3. Open a report as admin, change the status to Deleted, and verify that: • The status is updated. • The resident receives a notification.	Yes	

		reward points to the report owner (only once per report).	• No points are awarded. 4. Attempt to update a report that is already Completed or Deleted, and verify that the system prevents further updates.		
2	As a user, I want the website layout to be clear and easy to navigate, So that I can find features quickly without confusion.	• The main navigation menu shall include clearly labeled buttons (Submit Report, My Reports, Notifications, Rewards, Logout). • The user shall be able to access major features within 2-3 clicks. • The interface shall not	1. User opens the system and views the navigation menu. 2. Verify that all required navigation options are clearly labeled. 3. User accesses each main feature and verifies it can be reached within 2–3 clicks. 4. Verify that there are no unnecessary or duplicate options.	Yes	

		contain unnecessary or duplicate options.			
2	As a user, I want the system to respond quickly when I use its features so that I do not experience delays.	• The system shall load the user dashboard within 3 seconds under normal operating conditions. • The system shall display the list of reports within 3 seconds. • The system shall process report submission within 5 seconds.	1. User logs in and measures dashboard loading time. 2. User navigates to "My Reports" and observes loading time. 3. User submits a report and observes response time. 4. Verify that all actions are completed within the specified time limits.	Yes	

5.2 Integration Testing

In this section, integration testing is conducted to evaluate how different components of the system interact when combined together. The purpose of this testing is to ensure that all modules function correctly as a unified system and that data flows seamlessly between them.

Integration testing focuses on identifying issues related to communication between system components. It is performed after individual units have been tested and verified, ensuring that the

integration process does not introduce new errors. This testing helps confirm that the system operates as intended and delivers a consistent and reliable user experience.

5.2.1 Integration Hierarchy

The integration hierarchy illustrates the sequence and structure in which system components are combined and tested. It provides a clear view of how different modules interact and the order of execution between them. This helps in understanding the system flow and ensures that dependencies between components are properly managed during integration.

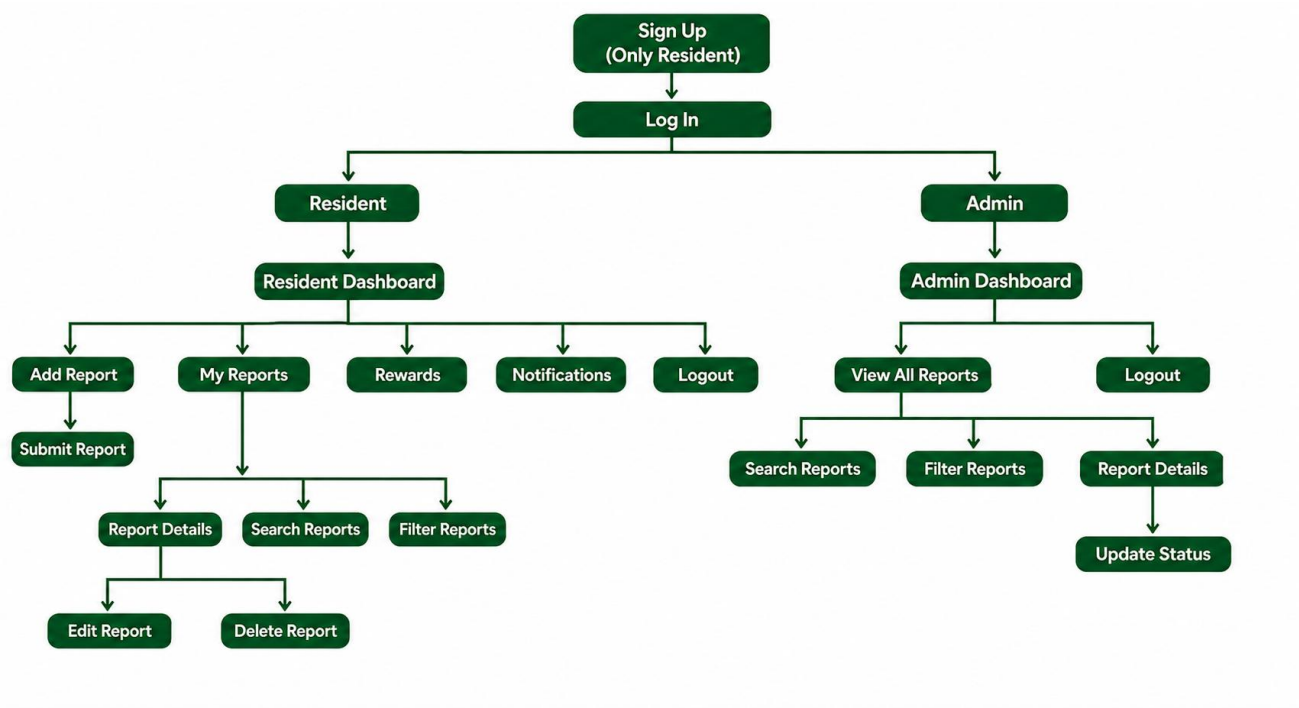


Figure 14: Navigation diagram for Rasheed system

5.2.2 Integration Testing Plane

In this sub-section, we present the integration testing plan table, which is designed to ensure that system components interact correctly with newly integrated components. This is achieved by examining a set of relevant test cases and indicating whether each interaction passes or fails.

Table 9: Testing plan table

System components	New component	Test case	Pass?	Comment
- Logging in the system	Viewing My Reports	1. A user was logged in the system successfully. 2. The logged in user selected the My Reports option.	Yes	
- Logging in the system - Viewing My Reports	Viewing Report Details	1. A user was logged in the system successfully. 2. The logged in user selected the My Reports option. 3. The logged in user selected a report from the list.	Yes	
- Logging in the system - Viewing My Reports - Viewing Report Details	Deleting a report	1. A user was logged in the system successfully 2. The logged in user selected the My Reports option. 3. The logged in user selected a report from the list. 4. The logged in user selected the Delete	Yes	

		Report option.		
- Logging in the system	Viewing Notifications	1. A user was logged in the system successfully. 2. The logged in user selected the Notifications option.	Yes	
- Logging in the system - Viewing Notifications	Displaying notification status	1. A user was logged in the system successfully. 2. The logged in user selected the Notifications option. 3. The system displayed notifications with their correct status .	Yes	

5.3 User Acceptance Testing

User Acceptance Testing (UAT) represents the final evaluation phase of the Rasheed system, where real users assess whether the system meets their practical needs and supports its intended purpose. In this section, the system is evaluated based on actual user interaction to determine its usability, functionality, and overall effectiveness in addressing real-world reporting scenarios.

The main objective of this testing is to examine how easily users can interact with the system's core features, including submitting reports, tracking their status, viewing rewards, and receiving

notifications. By observing user behavior and collecting feedback, this phase aims to verify whether the system delivers a smooth and intuitive user experience while meeting the defined requirements.

To conduct the testing, a group of participants was selected from the target user population. Each participant was asked to independently interact with the system and complete a set of essential tasks that reflect typical user activities. This approach ensures that the evaluation is based on realistic usage conditions and provides meaningful insights into the system's performance.

After completing the assigned tasks, participants were required to fill out a structured questionnaire designed to capture their feedback. The questionnaire consisted of 11 statements evaluated using a 5-point Likert scale, where 1 represents "Strongly Disagree" and 5 represents "Strongly Agree." The questionnaire was distributed using Google Forms, allowing responses to be collected efficiently and analyzed systematically.

The results obtained from this testing phase are presented in the following subsections. These include an overview of participant demographics, followed by a detailed analysis of questionnaire responses, which together provide a comprehensive evaluation of the system's usability, performance, and overall user satisfaction.

5.3.1 Demographics of Participants

User Acceptance Testing was conducted with five participants representing the target users of the Rasheed system. The participants were selected to reflect the characteristics of the intended user base, particularly residents in Saudi Arabia who are likely to encounter and report water and electricity issues.

Figure 15: Age distribution

In terms of age distribution, 40% of participants were between 18–24 years old, 40% were between 25–34 years old, and 20% were between 12–17 years old. This shows that the majority of participants fall within young adult age groups, which aligns with the expected primary users of the system.

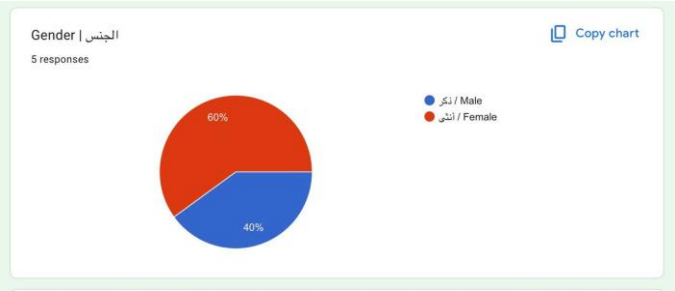


Figure 16: Gender Distribution

Regarding gender, 60% of participants were female, while 40% were male, providing a balanced representation. In terms of technical experience, 60% of participants reported having advanced technical skills, while 20% had intermediate experience and 20% had basic experience. This variation ensured that the system was evaluated from different usability perspectives.

Figure 17: Technical Experience

All participants had basic digital literacy skills, including the ability to use smartphones and web applications. While most participants were familiar with online platforms, they did not necessarily have prior experience with reporting utility-related issues, which aligns with the system's goal of being

accessible to everyday users. Overall, the diversity in demographics helped ensure that the system was tested under realistic conditions and provided meaningful feedback on usability and user experience.

5.3.2 Questionnaire/Interview Result

In this section, the results of the User Acceptance Testing (UAT) questionnaire are presented and analyzed. The purpose of this analysis is to evaluate the usability, performance, and overall user experience of the Rasheed system based on participants' feedback. Each result is illustrated using a chart, followed by a brief interpretation.

1) System Interface Attractiveness

The results indicate that 60% of participants rated the interface as highly appealing (4), while 40% gave the highest rating (5). This suggests that the system's visual design is well-received and meets user expectations in terms of attractiveness.

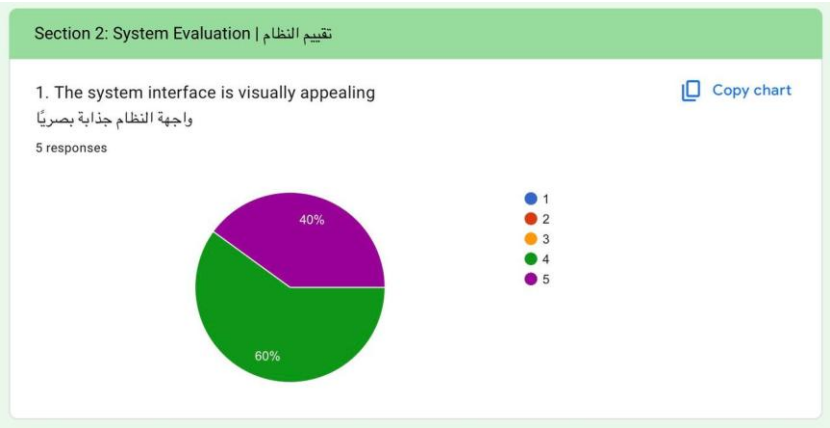


Figure 18: User ratings for system interface attractiveness

2) Ease of Navigation

A strong majority of participants (80%) rated navigation as very easy (5), while 20% rated it as easy (4). This demonstrates that users can move through the system smoothly without confusion.

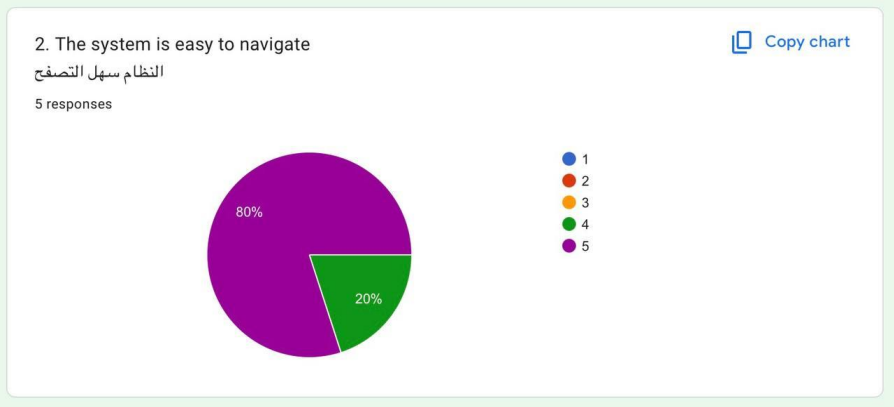


Figure 19:User ratings for ease of navigation

3) Clarity of Account Creation and Login

The results show that 80% of participants strongly agreed that the account creation and login process is clear and easy, while 20% agreed. This indicates that the authentication process is straightforward and user-friendly.

Figure 20:User ratings for account creation and login process

4) Ease of Submitting Reports

Most participants (60%) strongly agreed that submitting a report is straightforward, while 20% rated it as easy (4) and 20% gave a neutral rating (3). This suggests that although the process is generally simple, minor improvements could enhance clarity for some users.

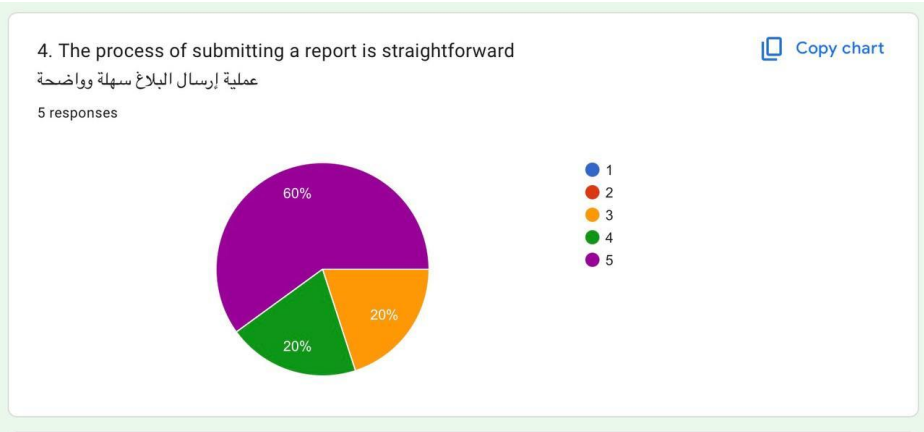


Figure 21:User ratings for report submission process

5) Tracking Report Status

The results show that 60% of participants strongly agreed that tracking report status is easy, while 40% agreed. This indicates that the system effectively supports users in monitoring their reports.

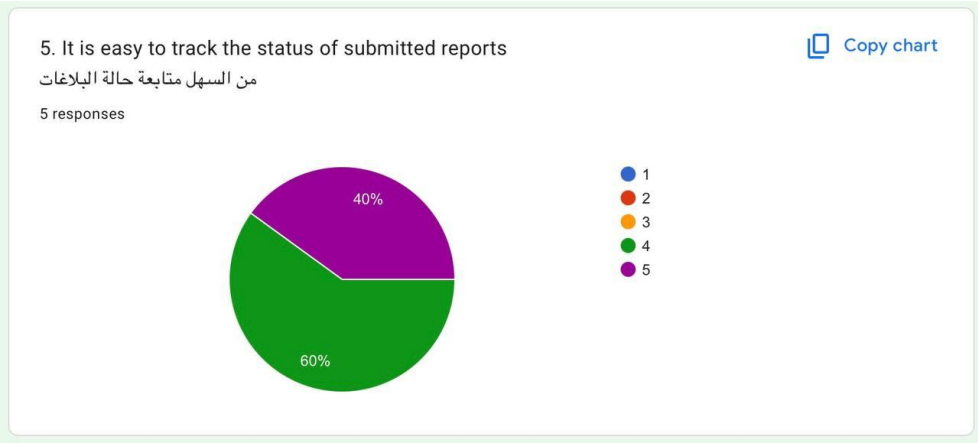


Figure 22:User ratings for tracking report status

6) Usefulness of Notifications

All participants (100%) strongly agreed that notifications provide useful updates about report status. This highlights the effectiveness of the notification feature in keeping users informed.

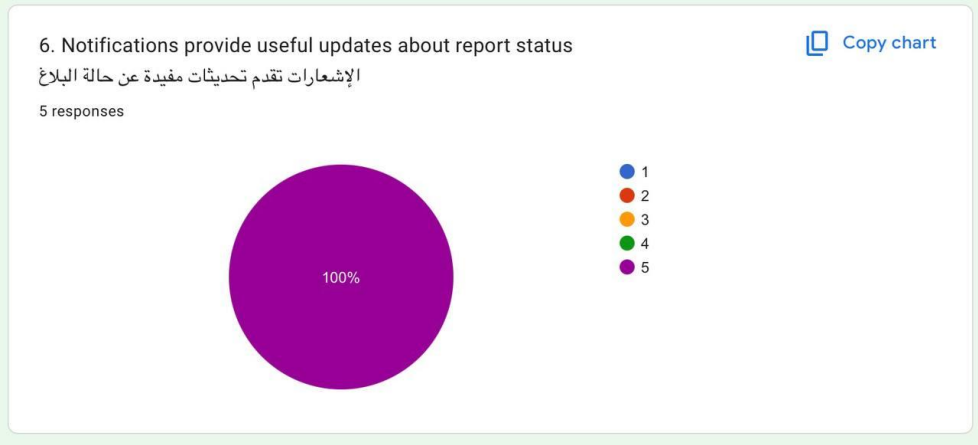


Figure 23:User ratings for usefulness of notifications

7) Clarity of Rewards System

The rewards system received strong feedback, with 60% of participants strongly agreeing that it is clear, while 20% agreed and 20% were neutral. This suggests that the feature is generally understandable but could benefit from slight improvements in explanation or presentation.

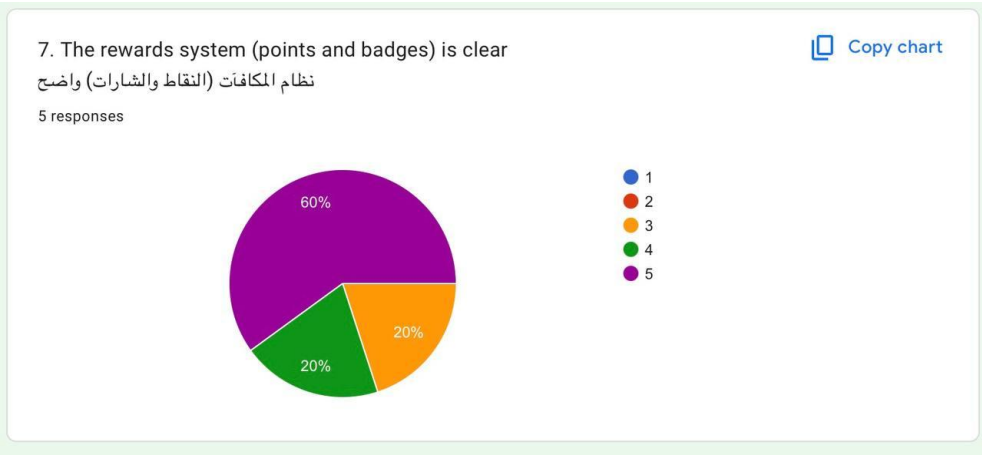


Figure 24:User ratings for rewards system clarity

8) System Responsiveness

A majority of participants (80%) strongly agreed that the system responds quickly to their actions, while 20% agreed. This indicates that performance requirements are successfully met.

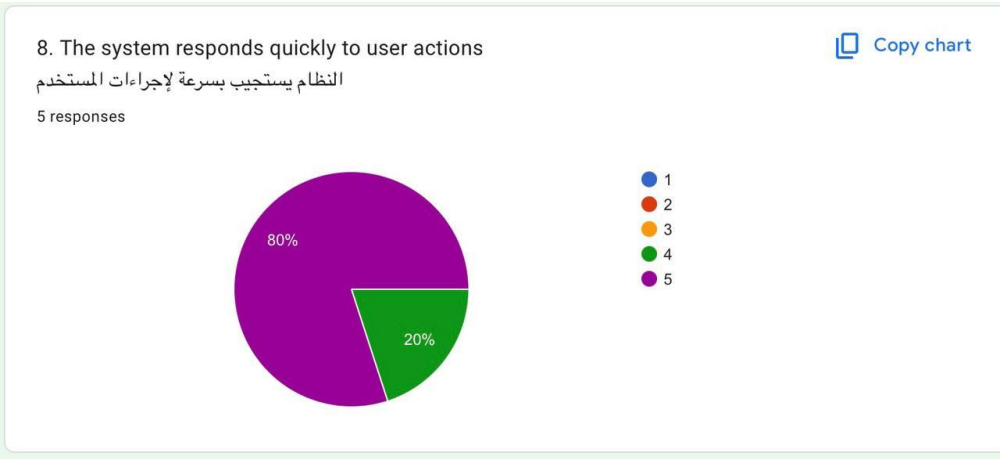


Figure 25:User ratings for system responsiveness

9) Design Consistency

The results show that 80% of participants strongly agreed that the system's design and layout are consistent, while 20% agreed. This reflects a well-structured and uniform interface across different pages.

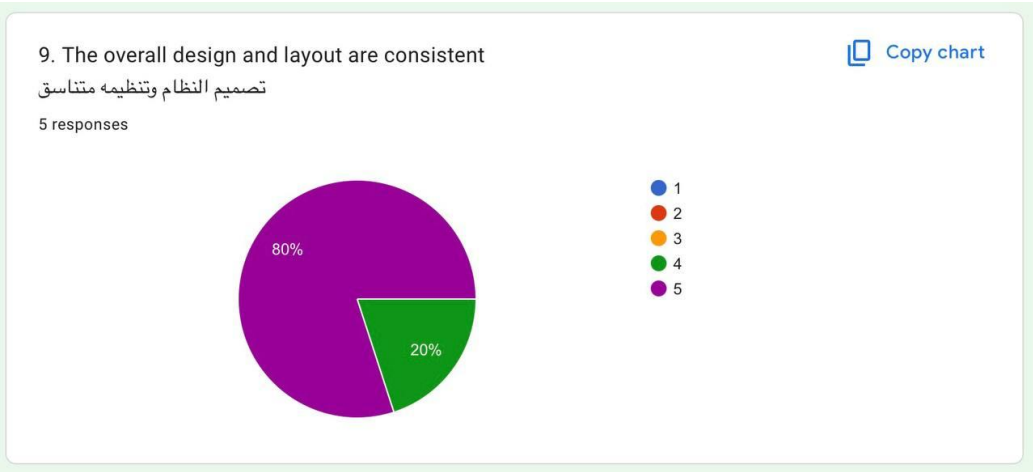


Figure 26:User ratings for design consistency

10) Willingness to Use the System

Most participants (60%) strongly agreed that they would consider using the system in real life, while 40% agreed. This demonstrates a high level of user acceptance and perceived usefulness.

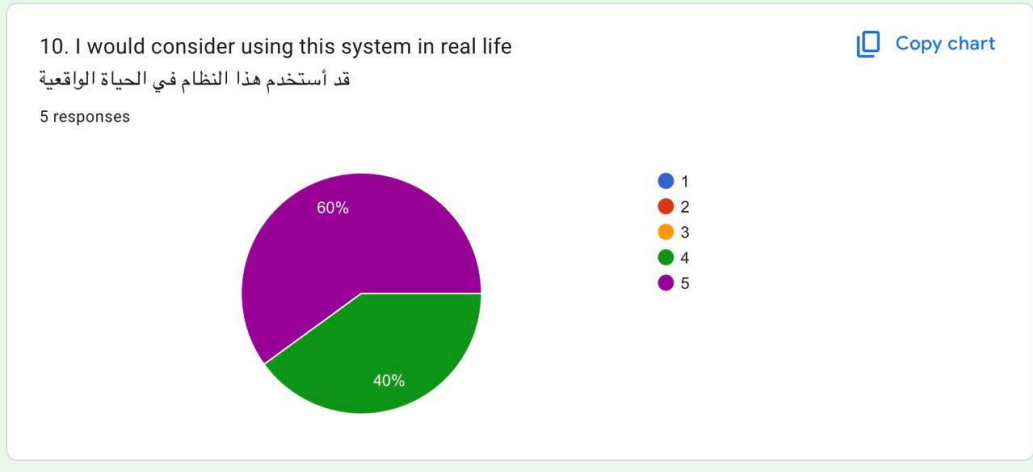


Figure 27:User ratings for willingness to use the system

11) Overall Satisfaction

The majority of participants (80%) reported being highly satisfied with their experience (5), while 20% rated it as 4. This confirms that the overall user experience of the system is positive.

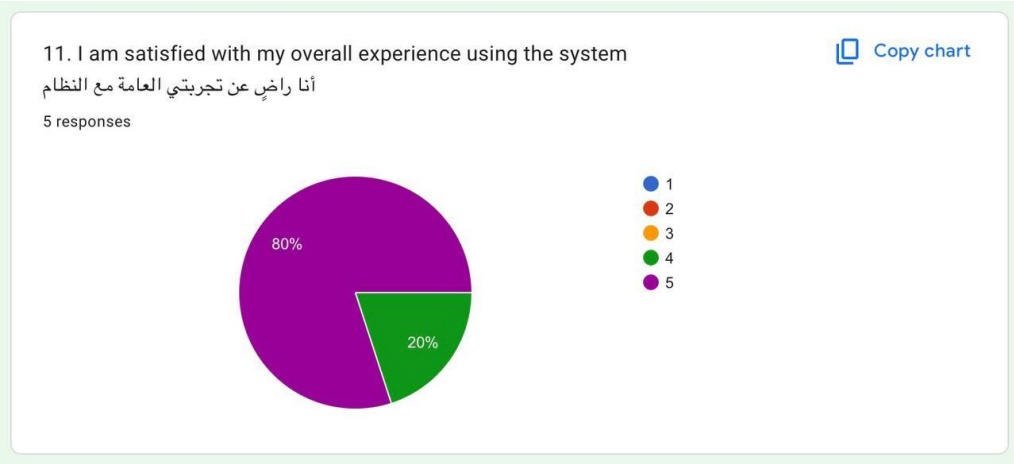


Figure 28:User ratings for overall satisfaction

Detailed questionnaire questions, full response results, and additional graphs are provided in Appendix C.

5.4 Discussion

The results of the User Acceptance Testing (UAT) indicate that the Rasheed system successfully meets its primary usability and functionality objectives. The consistently high ratings across key aspects such as ease of navigation, interface clarity, and overall satisfaction demonstrate that users were able to interact with the system effectively and without difficulty. This confirms that the system aligns well with its intended purpose and target users.

In addition, the questionnaire results highlight that features such as notifications and report tracking were particularly well-received. The strong agreement among participants shows that users value real-time updates and the ability to monitor the status of their reports, which enhances transparency and trust in the system. The rewards system also received positive feedback, indicating that it is clear and contributes to motivating user engagement.

From a non-functional requirements (NFR) perspective, the system performed well in terms of usability, responsiveness, and consistency. Participants reported that the system responds quickly to their actions and maintains a consistent design across different pages, which reflects good implementation of performance and usability requirements. These results suggest that the system not only functions correctly but also provides a smooth and efficient user experience.

However, some moderate ratings were observed in the report submission process. This suggests that certain steps within this feature may not be as intuitive as expected, possibly due to input complexity or lack of guidance. This highlights an area for improvement, where simplifying the workflow or enhancing form design could further improve usability.

Overall, the evaluation results can be considered positive. The majority of participants expressed satisfaction with the system and indicated a willingness to use it in real-life scenarios. These findings confirm that the design and development decisions were largely effective. With minor refinements, particularly in improving the report submission process, the Rasheed system has strong potential to serve as a reliable and user-friendly platform for reporting and managing utility issues.

6 Chapter 6: Conclusion and Future Work

6.1 Conclusion

The increasing pressure on water and electricity resources, particularly in countries such as Saudi Arabia, highlights the urgent need for more efficient and sustainable management approaches. As discussed in the introduction, factors such as population growth, urban expansion, and reliance on resource-intensive systems like desalination have made it essential to improve how these resources are monitored and managed. Addressing these challenges requires not only policy-level strategies but also practical technological solutions that support real-time communication and informed decision-making.

In response to this need, the Rasheed system was developed as a web-based platform that enables residents to report water and electricity issues in their communities. The system facilitates communication between users and administrators, allowing issues to be reported, tracked, and resolved efficiently. In addition, the inclusion of features such as notifications and a reward mechanism encourages user engagement and promotes community participation in sustainability efforts.

Throughout the development process, the project followed a structured software engineering approach, including requirements analysis, system design, implementation, and testing. User Acceptance Testing results indicated that the system is generally easy to use, visually clear, and effective in supporting its intended functions. Participants expressed positive feedback regarding usability, navigation, and responsiveness, while also highlighting minor areas for improvement.

Overall, the Rasheed system demonstrates how digital platforms can contribute to sustainable resource management by improving transparency, encouraging community involvement, and supporting faster response to infrastructure-related issues. The project aligns with national sustainability goals and showcases the potential impact of technology in addressing real-world environmental challenges.

6.2 Global and Local Impact

At the local level, the Rasheed system supports communities in Saudi Arabia by providing an accessible platform for reporting and tracking utility issues. This contributes to improved service efficiency, reduced resource waste, and increased public awareness of sustainability practices.

At the global level, the system reflects a broader trend toward using technology to address environmental and infrastructure challenges. Similar platforms can be adapted in different regions to support sustainable development goals, particularly those related to clean water, energy efficiency, and smart city initiatives.

6.3 Problems and Challenges

During the development of the Rasheed system, several technical and coordination challenges were encountered. One of the primary difficulties was related to database integration and back-end development. Establishing a reliable connection between the front-end interfaces and the database required careful handling of data flow, query execution, and error management. Challenges such as maintaining consistent data updates and ensuring correct synchronization between system components required additional debugging and testing.

Furthermore, integrating multiple functionalities—including report management, notifications, and the rewards system within a unified back-end structure required careful design to ensure smooth interaction between modules. Resolving logical dependencies between these components was a key challenge during implementation.

Another challenge involved coordinating work among team members, particularly when different parts of the system were being developed simultaneously. Ensuring compatibility between front-end and back-end components required continuous communication and integration efforts.

6.4 Limitations of the System

Despite its functionality, the system has some limitations. The platform currently supports a limited number of users and lacks advanced features such as real-time notifications using external services or integration with official utility providers.

Additionally, the evaluation of the system was based on a small sample size, which may not fully represent all potential users. The system interface is also currently available only in English, which may limit accessibility for some users.

6.5 Main Contribution of the Project

The main contribution of this project is the development of a functional prototype that demonstrates how a digital reporting platform can support sustainable management of water and electricity resources.

The system provides a structured and user-friendly way for residents to participate in identifying and resolving community issues.

Furthermore, the project highlights the importance of integrating user-centered design with sustainability goals, offering a practical example of how technology can bridge the gap between communities and service providers.

6.6 Future Work

Future improvements can enhance the system's functionality and impact. These include adding support for Arabic language to improve accessibility, integrating real-time notification systems such as SMS or email alerts, and connecting the platform with official utility services for direct report handling.

Additional features such as location-based reporting, advanced data analytics, and mobile application support can further improve user experience and system efficiency. Expanding testing with a larger and more diverse group of users would also provide more comprehensive feedback for system refinement.

7 References

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8 Appendix

Appendix A – Interview questions and transcripts:

A.1 Interview Questions:

1. Have you ever faced a water leak or electricity issue in your area? Did you report it?
2. What difficulties did you face while reporting the issue?
3. Have you ever decided not to report an issue? Why?
4. How important is it for you to track the status of your reported issue?
5. Would a reward system (points or badges) encourage you to report issues more often?
6. What features would make a reporting website easy and user-friendly for you?

A.2 Interview Transcripts:

Participant 1
Interviewer: Tala
Interviewee ID: P1
Age: 46
Gender: Male
Nationality: Saudi
Occupation: Doctor
Technical Experience: Moderate

Q1: Have you ever faced a water leak or electricity issue in your area? Did you report it?
Answer: Yes, I experienced a water leak in my area and attempted to report it.

Q2: What difficulties did you face while reporting the issue?
Answer: I did not know of any official website or platform where I could report the issue and be certain that someone would respond. I was unsure who to contact or how to ensure the problem would be handled. As a result, I reported it informally by informing the security guard at a nearby mosque.

Q3: Have you ever decided not to report an issue? Why?
Answer: I usually try to report issues whenever possible, even if I am uncertain whether someone will respond. I believe it is better to attempt reporting rather than ignore the problem.

Q4: How important is it for you to track the status of your reported issue?
Answer: It is very important for me to know what happens after I report an issue. I would like to receive confirmation that my report has been received and to know whether it is being worked on.

Q5: Would a reward system (points or badges) encourage you to report issues more often?

Answer: I would report issues regardless of whether a reward system exists. However, having points or badges would encourage me to report issues more quickly and follow up on them.

Q6: What features would make a reporting website easy and user-friendly for you?
Answer: I would prefer the website to be simple and easy to use. The layout should be clear and not crowded with too many buttons or options. I would also prefer logging in using my phone number or email instead of completing a lengthy registration process. After submitting a report, I would like to immediately receive confirmation that it has been received.

Participant 2
Interviewer: Lama
Interviewee ID: P2
Age: 27
Gender: Female
Nationality: Saudi
Occupation: Employee
Technical Experience: High

Q1: Have you ever faced a water leak or electricity issue in your area? Did you report it?
Answer: Yes, I have faced several issues and reported them.

Q2: What difficulties did you face while reporting the issue?
Answer: The process required entering information repeatedly each time. It was not straightforward to reach the specific reporting section, and the process involved too many inputs instead of quick access.

Q3: Have you ever decided not to report an issue? Why?
Answer: Yes, sometimes I chose not to report issues because the process was long and tracking the report was not easy.

Q4: How important is it for you to track the status of your reported issue?
Answer: It is very important.

Q5: Would a reward system (points or badges) encourage you to report issues more often?
Answer: Yes, it would encourage me.

Q6: What features would make a reporting website easy and user-friendly for you?
Answer: I would prefer quick access with minimal clicks, a chatbot for support, saved profile information to avoid repeated inputs, clear tracking stages, and notification updates.

Participant 3
Interviewer: Jood
Interviewee ID: P3
Age: 23
Gender: Male
Nationality: Saudi
Occupation: Employee
Technical Experience: High

Q1: Have you ever faced a water leak or electricity issue in your area? Did you report it?

Answer: Yes, we experienced a power outage, but I did not report it because I assumed someone else had already done so.

Q2: What difficulties did you face while reporting the issue?
Answer: The process was unclear, and I did not know which authority to contact. It also seemed time-consuming.

Q3: Have you ever decided not to report an issue? Why?
Answer: Yes, if the issue appears minor, I sometimes choose not to report it and assume it will be fixed without my involvement.

Q4: How important is it for you to track the status of your reported issue?
Answer: It is important for me to know whether the report has been received and whether progress is being made.

Q5: Would a reward system (points or badges) encourage you to report issues more often?
Answer: Rewards are not the main reason, but they would encourage greater participation.

Q6: What features would make a reporting website easy and user-friendly for you?
Answer: The website should be simple and fast, with clear instructions, a short form, and easy mobile access.

Participant 4
Interviewer: Shatha
Interviewee ID: P4
Age: 45
Gender: Female
Nationality: Saudi
Occupation: Employee
Technical Experience: Moderate

Q1: Have you ever faced a water leak or electricity issue in your area? Did you report it?
Answer: Yes, we had a water leak near our house. I reported it through an application because I was concerned about my family and did not want the issue to remain unresolved.

Q2: What difficulties did you face while reporting the issue?
Answer: The process felt longer than necessary. I had to re-enter my information each time, and it was sometimes unclear where to click to choose the issue type. I prefer simple and straightforward steps.

Q3: Have you ever decided not to report an issue? Why?
Answer: Yes, if the issue seems minor, I sometimes hesitate because reporting takes time. As a working mother, my time is limited, and I may assume someone else will report it.

Q4: How important is it for you to track the status of your reported issue?

Answer: It is very important. I want to know whether the issue has been addressed or if someone is handling it. Seeing updates provides reassurance.

Q5: Would a reward system (points or badges) encourage you to report issues more often?
Answer: Yes, even small incentives would make me feel appreciated and encourage me to participate more.

Q6: What features would make a reporting website easy and user-friendly for you?
Answer: I prefer clear and quick buttons with minimal typing. It would help if my information is saved after the first time. I would also like to receive notifications when the status changes and have access to quick help if needed.

Appendix B – Questionnaire Responses and graphs:

1. Age:
-12 – 18
- 19 – 30
- 31 – 45
- 46+

2. Gender:
- Male
- Female

3. How often do you experience water or electricity problems in your area?
Very often
- Sometimes
- Rarely
- Never

4. Have you reported a utility problem before?
-Yes
- No

5. If yes, how did you report it?
- Phone call
- Government website
- Social media
- I did not report

6. How easy was the reporting process?
- Very easy
- Easy
- Difficult
- Very difficult

7. How long does it usually take to resolve utility issues in your area?

- Very quickly
- Within a few days
- Takes long
- Not resolved

8. Would you use a website to report water or electricity issues?
- Yes
- No
- Maybe

9. Do you think attaching a photo improves the quality of a report?
- Yes
- No

10. Is selecting a severity level important when reporting a problem?
- Yes
- No

11. Would receiving a badge after multiple valid reports motivate you?
- Yes
- No

12. Would you recommend such a reporting platform to others?
- Yes
- No

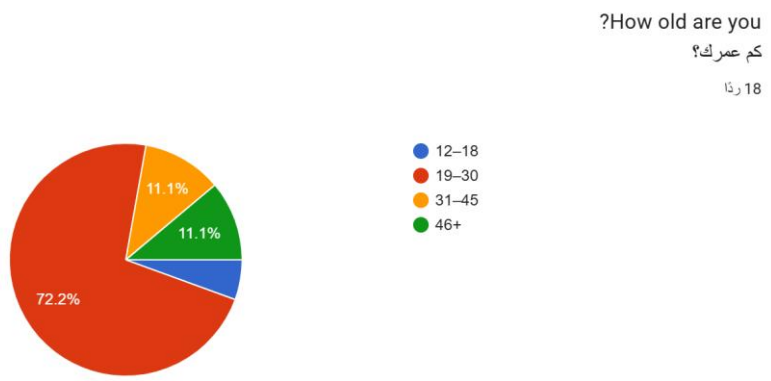


Figure 29:Questionnaire responses -Question1.

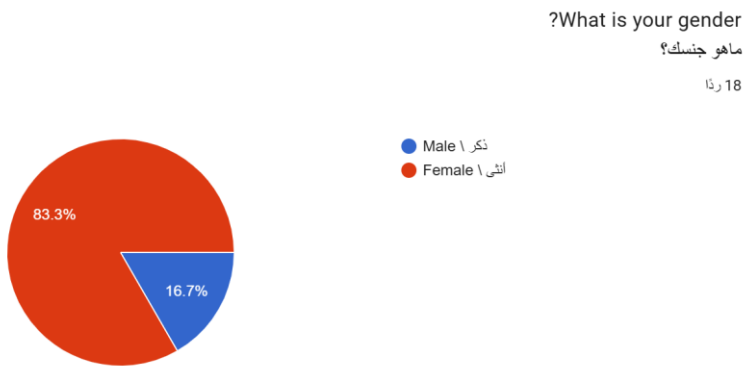


Figure 30:Questionnaire responses -Question2.

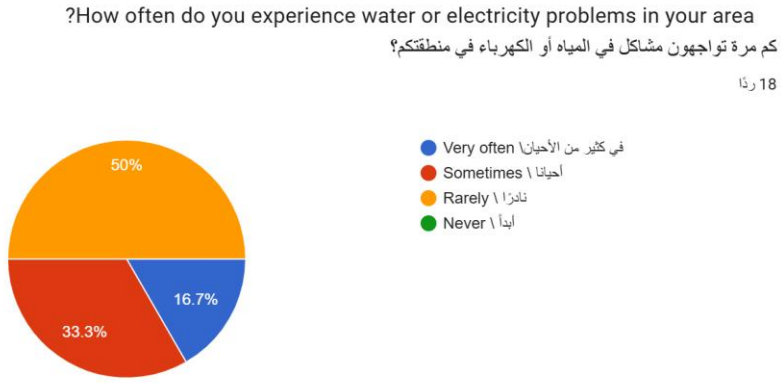


Figure 31:Questionnaire responses -Question3.

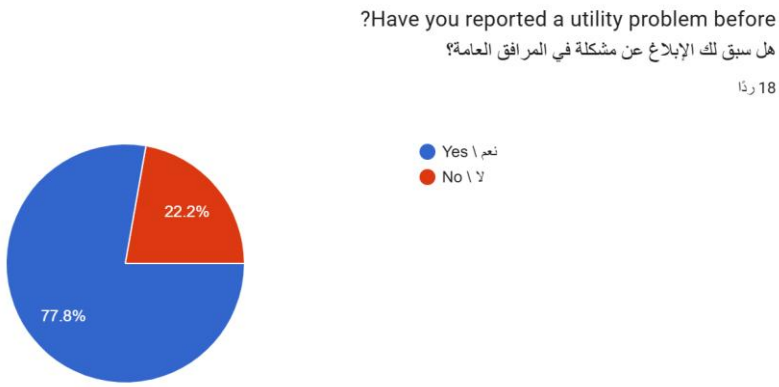


Figure 32:Questionnaire responses -Question4.

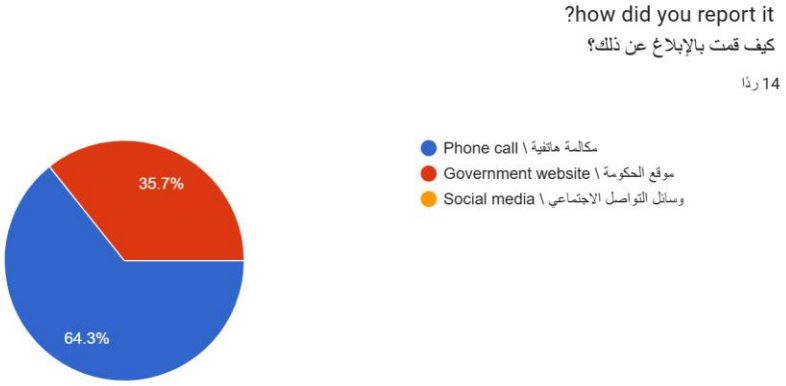


Figure 33:Questionnaire responses -Question5.

Figure 34:Questionnaire responses -Question6.

Figure 35:Questionnaire responses -Question7.

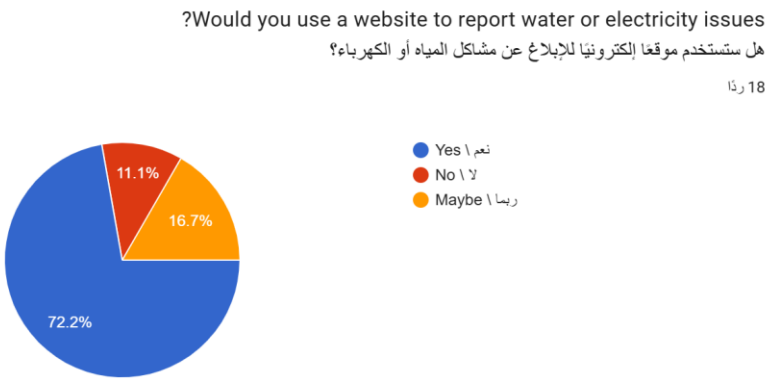


Figure 36:Questionnaire responses -Question8.

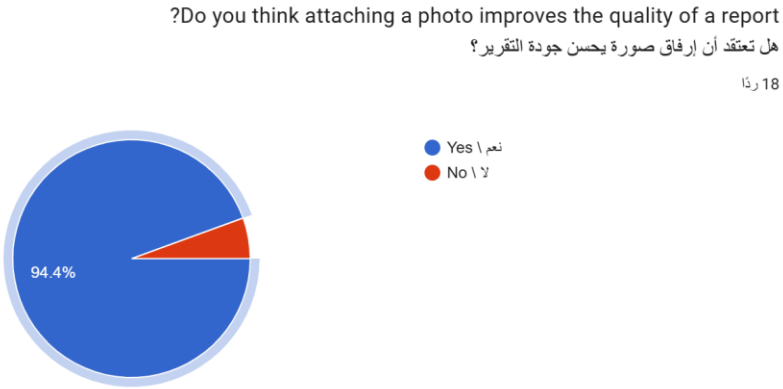


Figure 37:Questionnaire responses -Question9.

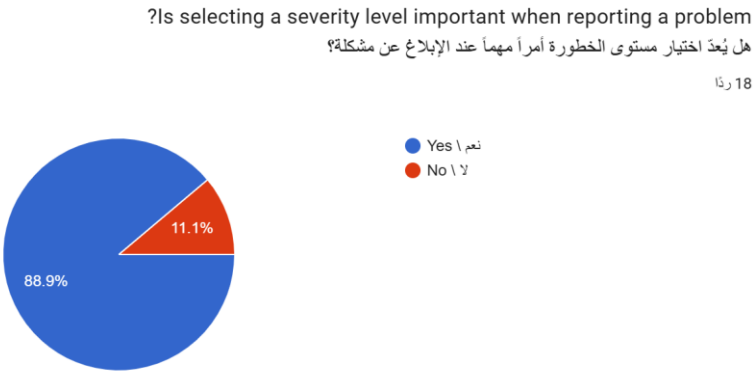


Figure 38:Questionnaire responses -Question10.

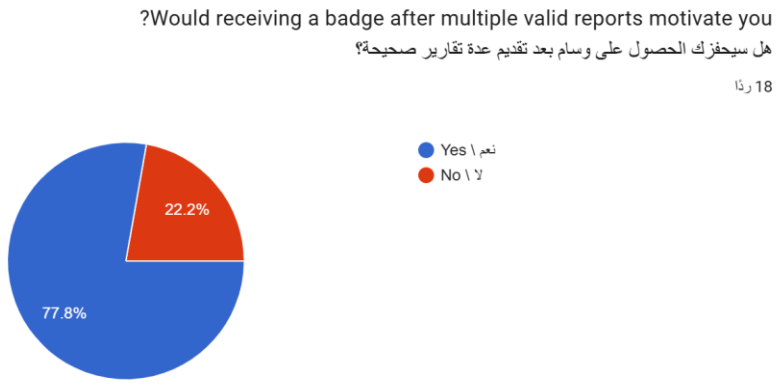


Figure 39:Questionnaire responses -Question11.

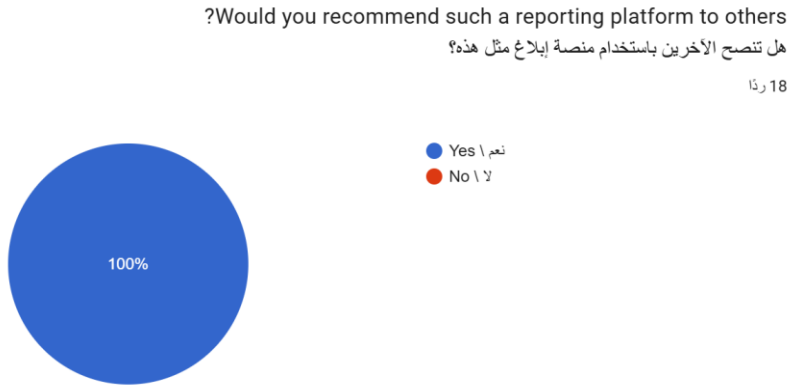


Figure 40:Questionnaire responses -Question12.

Form_Responses

طابع زمني	How old are you? كم عمرك؟	What is your gender? ماهو جنسك؟	?How often do you experience water or electı كم مرة تواجهون مشاكل في المياه أو الكهرباء في منطقتكم؟	?Have you reported a utility problem before هل سبق لك الإبلاغ عن مشكلة في المرافق العامة؟	?how did you report it كيف قمت بالإبلاغ عن ذلك؟	?How ea ية الإبلاغ؟
10:52:08 2026/02/26 ص	19–30	Female \ أنثى	Sometimes \ أحيانًا	Yes \ نعم	Government website \ موقع الـ	
3:33:09 2026/02/27 ص	19–30	Female \ أنثى	Sometimes \ أحيانًا	Yes \ نعم	Phone call \ مكالمة هاتفية	
4:44:00 2026/02/27 ص	12–18	Female \ أنثى	Rarely \ نادرًا	No \ لا		
5:38:17 2026/02/27 ص	19–30	Female \ أنثى	Very often \في كثير من الأحيان	Yes \ نعم	Phone call \ مكالمة هاتفية	
6:36:58 2026/02/27 ص	19–30	Female \ أنثى	Rarely \ نادرًا	No \ لا		
6:51:50 2026/02/27 ص	19–30	Male \ ذكر	Sometimes \ أحيانًا	Yes \ نعم	Government website \ موقع الـ	
11:53:02 2026/02/27 ص	19–30	Female \ أنثى	Sometimes \ أحيانًا	Yes \ نعم	Phone call \ مكالمة هاتفية	
12:51:02 2026/02/27 م	19–30	Female \ أنثى	Very often \في كثير من الأحيان	Yes \ نعم	Phone call \ مكالمة هاتفية	
7:59:06 2026/02/27 م	19–30	Female \ أنثى	Very often \في كثير من الأحيان	Yes \ نعم	Government website \ موقع الـ	
11:14:44 2026/02/27 م	31–45	Male \ ذكر	Rarely \ نادرًا	Yes \ نعم	Government website \ موقع الـ	
11:15:14 2026/02/27 م	31–45	Female \ أنثى	Rarely \ نادرًا	No \ لا		
11:15:43 2026/02/27 م	19–30	Female \ أنثى	Rarely \ نادرًا	Yes \ نعم	Phone call \ مكالمة هاتفية	
11:16:51 2026/02/27 م	19–30	Female \ أنثى	Sometimes \ أحيانًا	Yes \ نعم	Government website \ موقع الـ	
11:18:13 2026/02/27 م	19–30	Male \ ذكر	Rarely \ نادرًا	Yes \ نعم	Phone call \ مكالمة هاتفية	
11:28:11 2026/02/27 م	19–30	Female \ أنثى	Rarely \ نادرًا	Yes \ نعم	Phone call \ مكالمة هاتفية	
11:30:41 2026/02/27 م	46+	Female \ أنثى	Rarely \ نادرًا	Yes \ نعم	Phone call \ مكالمة هاتفية	

Figure 41:Form_Responses

https://docs.google.com/spreadsheets/d/1wFivEtCF1CixOyVoxqjZ3TCU4i4uYDjV8zuM9W2PI5o/edit?usp=sharing

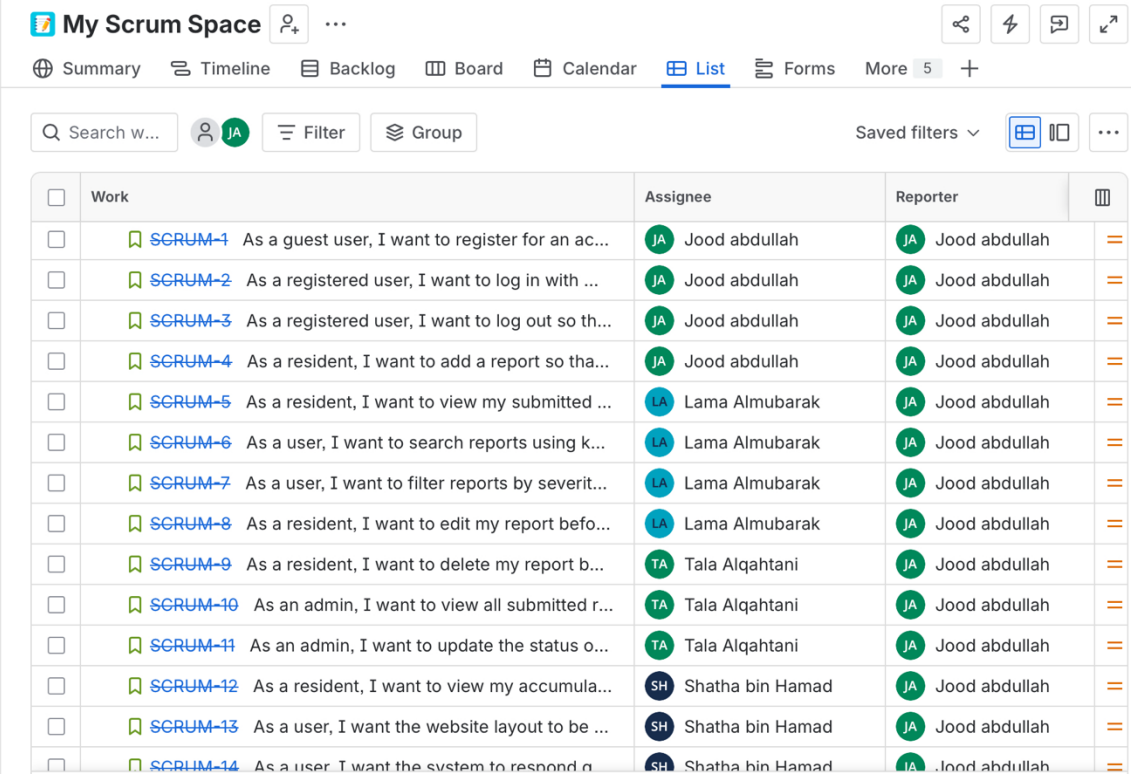
Appendix C – UAT Questionnaire (Questions, responses and graphs):

Participants were asked certain demographic questions, they were also asked to evaluate the system by responding to the following statements using a 5-point Likert scale (Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree):

1. The system interface is visually appealing.
2. The system is easy to navigate.
3. The process of creating an account and logging in is clear.
4. The process of submitting a report is straightforward.
5. It is easy to track the status of submitted reports.
6. Notifications provide useful updates about report status.
7. The rewards system (points and badges) is clear.
8. The system responds quickly to user actions.
9. The overall design and layout are consistent.
10. I would consider using this system in real-life situations.
11. I am satisfied with my overall experience using the system.

	A	B	C	D	E	F	G
	Form_Responses						
1	Timestamp	العمر | Age	الجنس | Gender	مستوى الخبرة التقنية | Technical Experience	1. The system interface is visually appealing واجهة النظام جذابة بصريًا	2. The system is easy to navigate النظام سهل التصفح	3. The process of creating an acc إنشاء الحساب وتسجيل الدخول واضحة وسهلة
2	28/04/2026 21:35:38	18–24	Female / أنثى	Advanced / متقدم	4	5	
3	28/04/2026 22:27:05	18–24	Male / ذكر	Advanced / متقدم	5	5	
4	28/04/2026 23:03:22	12-17	Female / أنثى	Intermediate / متوسط	5	4	
5	28/04/2026 23:03:42	25–34	Female / أنثى	Advanced / متقدم	4	5	
6	28/04/2026 23:13:10	25–34	Male / ذكر	Basic / مبتدئ	4	5	

Figure 42: Form Responses 2

Form Responses link:
https://docs.google.com/spreadsheets/d/1xrNMCL1rSVZBxhj3DNpHRWLP9lK5uzRCqQ2vo1yBP_M/edit?usp=sharing

Jira Screenshot: